Lecture Transcript by Slide

Week 4 · Chapter 3: Analyzing EKG Rhythm Strips, with a 12-lead review · 1 h 10 min (from 33:18 of the Week 3 recording)

Auto-transcribed, so expect occasional errors, especially in technical terms. Timestamps are positions in the recording.

0:33:18 Chapter 3: Analyzing EKG Rhythm Strips (title slide)

0:33:47 To Analyze EKG Rhythm Strip

0:34:48 Systematic Approach to Arrhythmia Interpretation

0:35:37 Practice Strip 3.1

0:41:12 Figuring Rates Based on the Number of QRS Complexes in a 6-Second Strip

0:42:45 Systematic Approach to Arrhythmia Interpretation (revisited)

0:46:42 Calculating Heart Rate – Regular Rhythms

0:47:26 Practice Strip 3.1: Rate

0:48:43 Calculating Heart Rate – Regular Rhythms (revisited)

0:49:38 Calculating Heart Rates: Methods

0:49:53 Figuring Rates in a 6-Second Strip (revisited)

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0:51:47 P Waves

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0:55:00 PR Interval

0:55:20 Practice Strip 3.1: PR Interval

0:56:30 PR Interval (revisited)

0:56:58 QRS Complex

0:57:25 Practice Strip 3.1: QRS Complex

0:58:56 QRS Complex (revisited)

0:59:12 Role of the AV Node

0:59:32 Ventricular vs. Supraventricular

1:00:35 Electrical Conduction System (PowerPoint editor view)

1:05:40 Ventricular vs. Supraventricular (revisited)

1:06:57 Width of QRS Complex

1:10:33 Chapter 3 Practice Strips

1:12:28 Appendix C: 12-Lead Electrocardiography (title slide)

1:12:52 12 Lead Electrocardiogram

1:13:32 Indications for a 12-lead EKG

1:19:19 Chest Lead Placement

1:19:33 Rule of Electrical Flow

1:19:47 Lead

1:20:04 Leads: Bipolar and Unipolar

1:20:35 Standard Limb Leads

1:21:18 Augmented Leads

1:27:09 Frontal Plane Leads (continued)

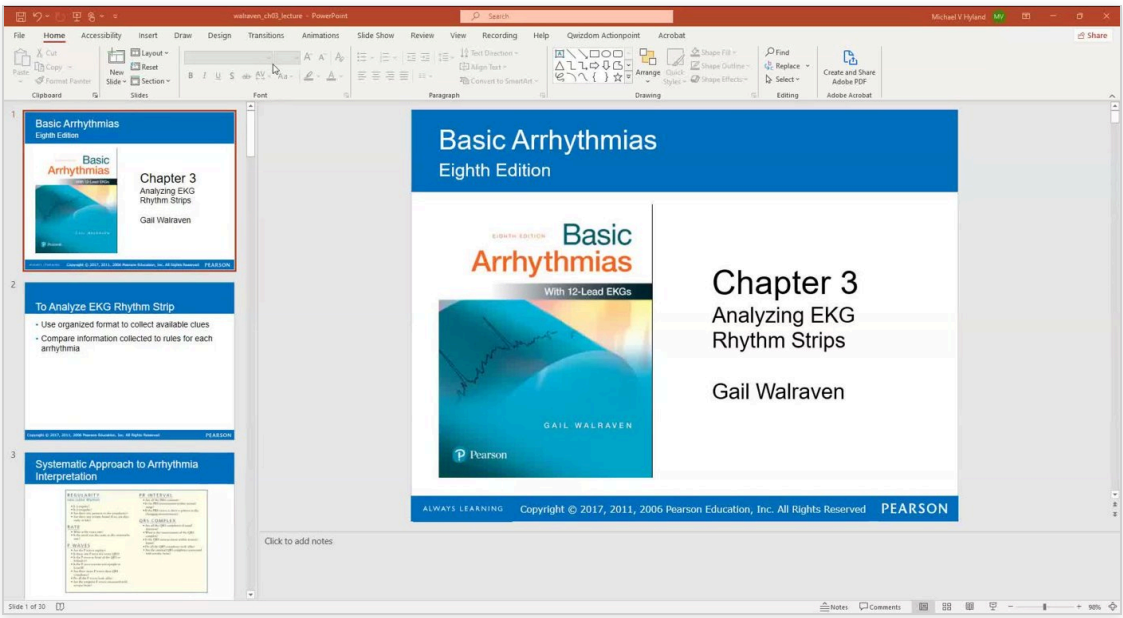
1:27:58 Horizontal Plane Leads

1:29:27 Standard Layout of 12-Lead EKG

1:33:00 Practice Strip 3.1

1 Chapter 3: Analyzing EKG Rhythm Strips (title slide)

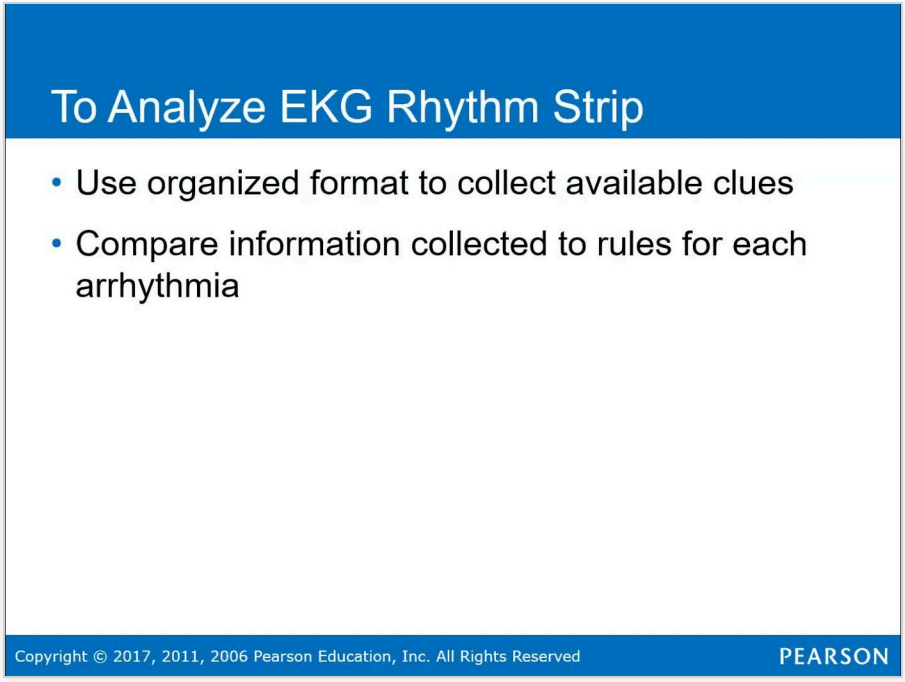
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2 To Analyze EKG Rhythm Strip

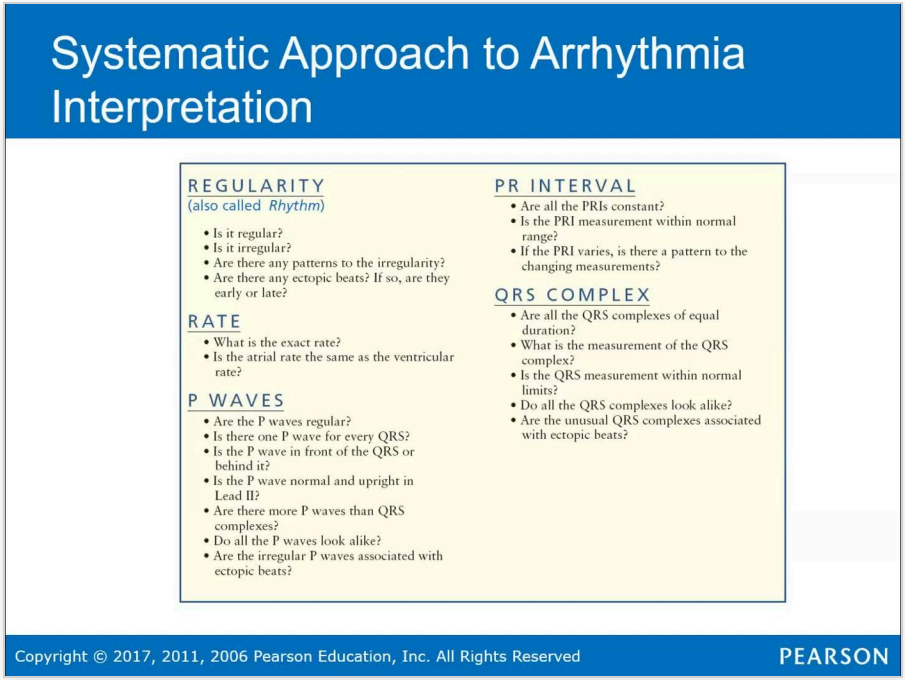
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3 Systematic Approach to Arrhythmia Interpretation

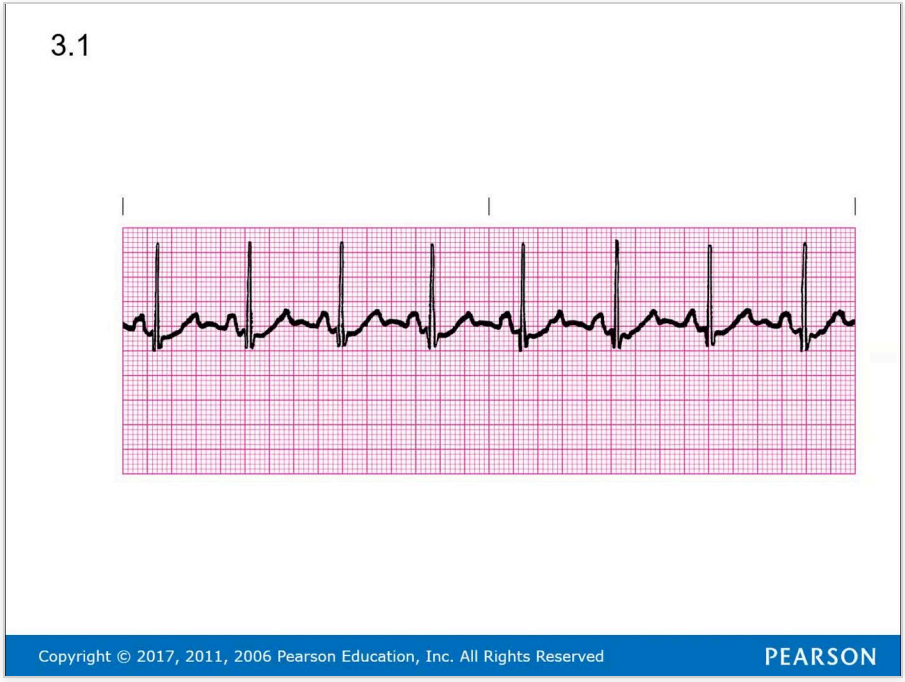
0:34:48 – 0:35:37



(Nothing was said while this slide was shown.)

4 Practice Strip 3.1

0:35:37 – 0:41:12



0:36:27 I just want to thank you so much for your time today. Thank you. Thank you so much. Thanks. We're excited to hear from you. Thank you. Thank you for coming today. Thank you for having me. I'm going to open it up. eight R-waves. So what we're going to do here is we're going to count the number of small boxes between these R-waves.

0:36:56 Okay, and I'm going to count fast, but if this is too fast for you, let me know. So I will tell you that this PowerPoint is online. It does have a lot of practice strips at the end of the PowerPoint. So although I didn't print it out for you, you do have access to this strip on the PowerPoint, which is in

0:37:21 and web-enhanced version of this. So I usually count in fives. So the large boxes, we know are fives, right? So I'm gonna go five, 10, 15, and then there's three here, one, two, three, 16, 17, 18, and then one there, 19. And then I'm gonna write the number 19 right up here at the top of the strip. strip. So I don't have to go back and count that again. I know there's 19 small boxes between the first and the second R wave on this strip. And then I'm going to do this across the strip. So this one starts right on a line. Thank you. Makes my counting a easier. 5, 10, 15, and then 4, make 19. 5, 10, 15, 16, 17, 18, and that one sits right in the middle. So I'm actually going to go 18.5 right there. So in terms of counting the small boxes, we're going to do holesies and handsies. Okay, so if the number falls in between, we're going to count it as a half. If the r-wave falls in between, we're going to count it as a

0:38:37 half, okay? If it falls on the line, we're going to count it as a whole, okay? And I'm going to make a comment about that in a second. Okay, five, ten, fifteen, sixteen, seventeen, eighteen and a half. Five, 10, 15, 16, 17, 18, 19, five, 10, 15, 16, 17, 18, 19, five, 10, 15, 16, 17, 18, 19 and a half. Everybody follow me. So I got 19, 19, 18 and a half, 18 and a half, 19, 19, 19 and a half. In terms of regularity, this rhythm is regular.

0:39:35 So what I'm going to say to you is very important. If the R wave, the difference between the R waves, between each set of R waves, so let's say we had 19, 19, 19, 20, 21, 22, right, that would be irregular.

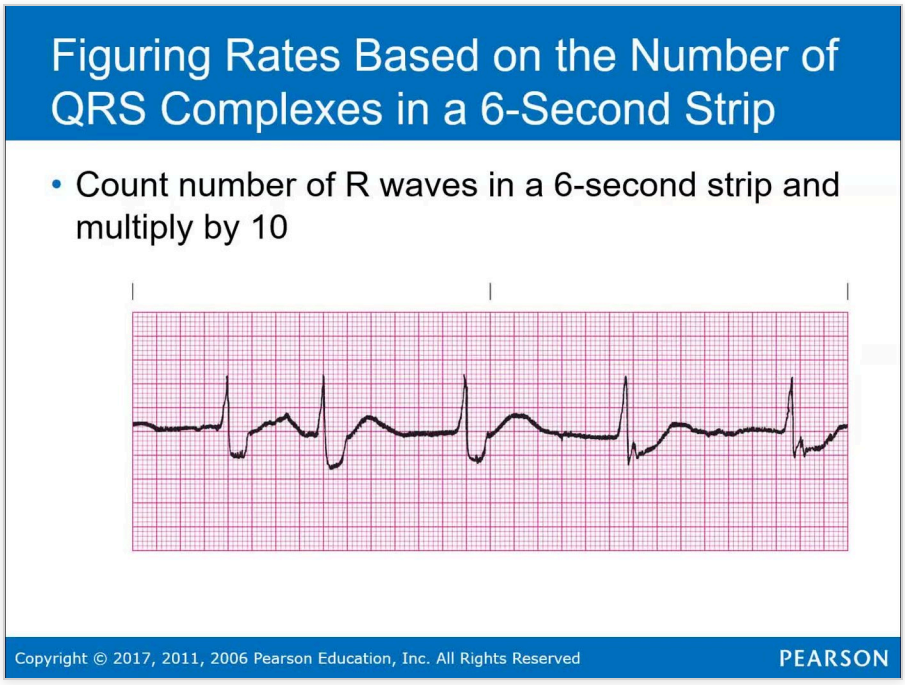
0:40:06 But, as long as the spaces, all the spaces, are within one and a half boxes of each other, so as long as all of the R-wave spacings across the strip are within one and a half boxes of each other, we are going to deem that rhythm regular.

0:40:32 Once the difference gets more than two boxes, or a box and a half, we're going to call it irregular. Everybody follow. It would be that difference of all boxes. When we have differences in the spacing from R to R across the strip greater than one and a half boxes, the rhythm is going to be irregular.

0:41:11 So then you guys do have this strip in your PowerPoint, so if we count the

5 Figuring Rates Based on the Number of QRS Complexes in a 6-Second Strip

0:41:12 – 0:42:45

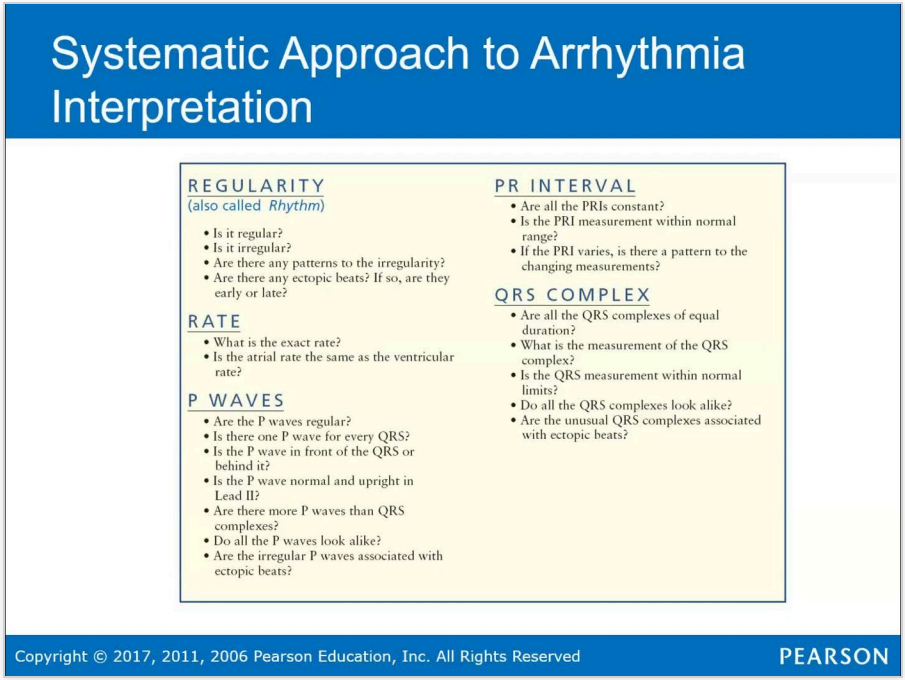


0:41:20 number of small boxes between these our waves 5, 10, 15, 20, 5, 10, 15, 20, 25, 29 and a half. Five, ten, fifteen, twenty, twenty-five, thirty, thirty-three, and then two halves make thirty-four. I got a half on each end. That's why I said two halves. Five, ten, fifteen, twenty, twenty-five, thirty, thirty-three, thirty-four, and two halves make thirty-five. So you can see the spacing between those R waves is completely irregular. So this rhythm will be deemed irregular. So the spacing of the heartbeat is irregular across the strip. So that means the heart's not beating at regular intervals is what that means. You have a regular wave and then it switches into an irregular

0:42:35 Absolutely. Correct. Correct. But that's all we have to go by. And then not to get overly confusing here. So the primarily the regularity is going to be regular, or it's going to be irregular. And then there's two other regularities that we can encounter.

6 Systematic Approach to Arrhythmia Interpretation (revisited)

0:42:45 – 0:46:42



0:43:07 So the first irregular that we can encounter is somebody has a regular rhythm. And remember now, last week I was talking about irritability. And I was talking about how irritability can cause a pacemaker cell to fire early or prematurely.

0:43:39 So some of these cardiac pacemaker cells can fire early and create one single beat. So the SA node is the primary pacemaker, which is firing at regular intervals, but now I get a spot in the AV node that gets irritable, and it fires an early beat before the SA node can fire again.

0:44:02 So I can have this regular rhythm interrupted by one early ectopic beat. In which case the rhythm itself will be regular, but it will be regular with one ectopic. Or sometimes in that six-second strip, that irritable spot in the AV note can fire twice or three times. So depending on how many irregular beats actually interrupt the regular rhythm. So it's typically going to be regular with one ectopic or regular with two ectopics or regular with three ectopics. And that will make more sense in a few weeks when we talk about atrial rhythms.

0:44:47 Okay. And then the last piece of the regularity that we may encounter, and we'll discuss this again at the end of the semester when we talk about heart blocks, but there's a regularity which is like an oxymoron. Regularly irregular. Yeah, I know, right? Doesn't make any sense. but neither does jumbo shrimp, right? We all like the really big shrimp, right? But we call them shrimp, right? So what happens in this one particular rhythm is you'll get three or four beats on the strip.

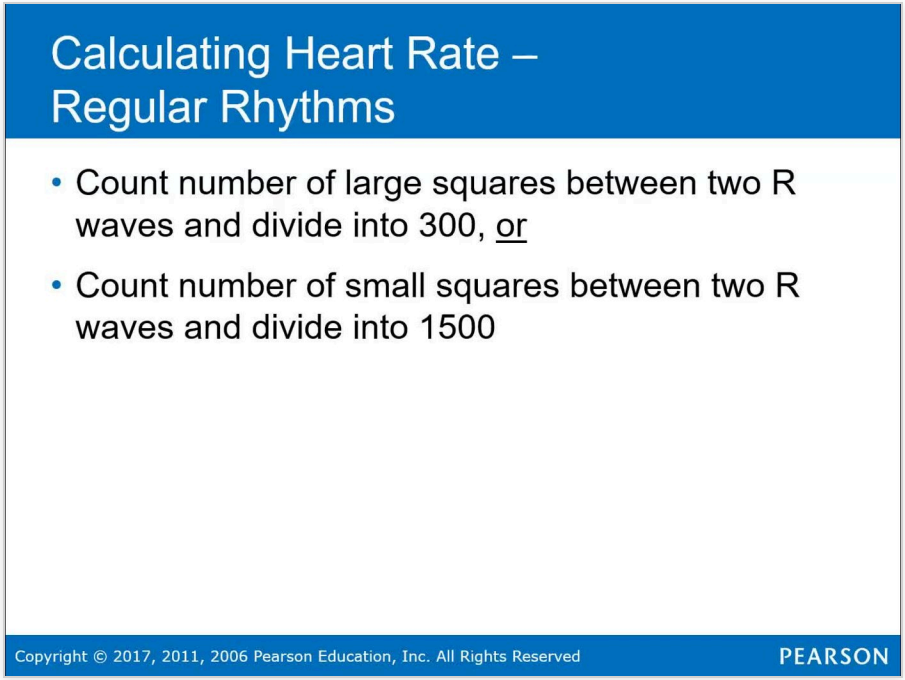
0:45:27 And the spacing between those three or four beats is irregular. So counting the small boxes between the first and second, second, third, third and fourth, those numbers are irregular, which makes that part of the rhythm irregular. But then what happens is, it plays those first four beats, then it plays those same four beats over again. So the rhythm is irregular, but it repeats that irregular pattern regularly.

0:45:56 You guys know what a record is? I'm showing my age here. But if you got a bad scratch on the record, right, it would play part of the song, and then it would skip back and play the same part of the song, and skip back and play the same part of the song. If your parents ever say to you, you sound like a broken record, that's what they mean because you're repeating yourself over and over and over again. This rhythm is like a broken record. So it's irregular, but it regularly repeats itself. Questions on regularity, that is going to be important.

0:46:31 The next thing we're going to talk about is heart rate. and there are going to be two methods to count the heart rate. You see that

7 Calculating Heart Rate – Regular Rhythms

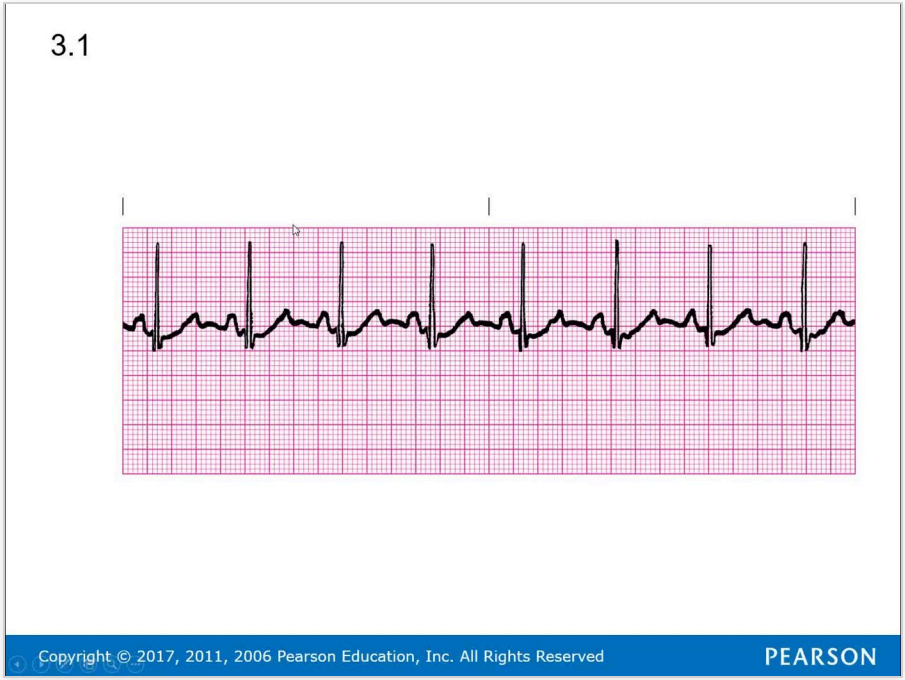
0:46:42 – 0:47:26



0:46:44 there's two methods on here. Cross out the first method that is listed. We are not going to use count the number of large squares between our waves and divide it into three hundred because counting a large box, well what if it's accurate way to calculate somebody's heart rate from an EKG strip is to count the number of small boxes between our waves and divide that number into 1500. So I'm going to go back to that same strip

8 Practice Strip 3.1: Rate

0:47:26 – 0:48:43

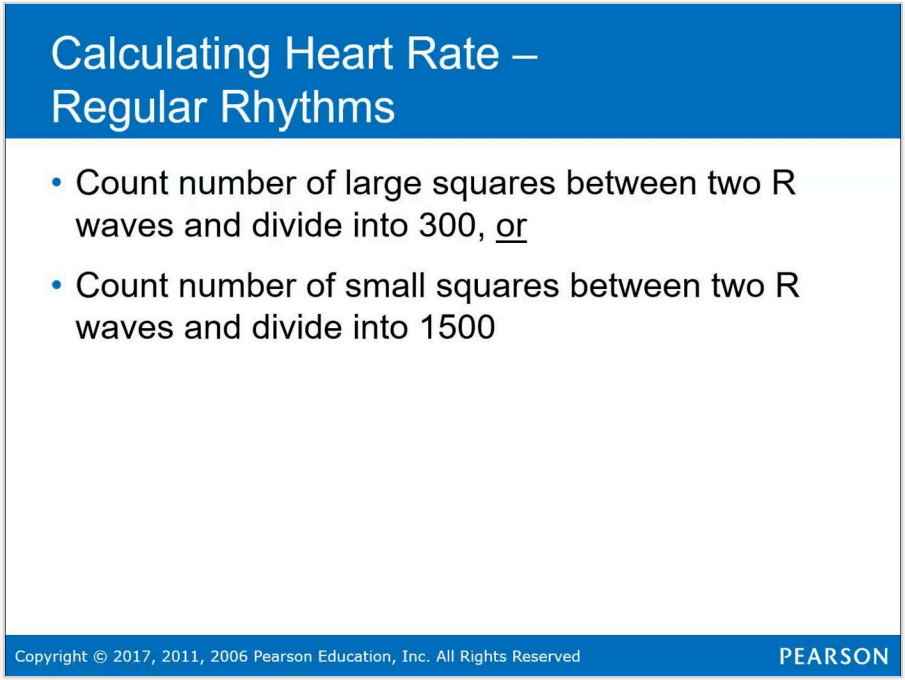


0:47:26 again and so I have if I remember correctly we have 19 19 18 and a half 18 and a half 19 19 19 and a half so typically you could average all that spacing out we said that this rhythm is regular I usually just go with majority wins right there's four spaces here that have 19 boxes. So I'm going to go 1500 divided by 19. I thought you guys were going to do that math in your head. What do you got Jess? 78.9. Okay, so the heart rate for this rhythm right here, and I would just round that up to 79. If it's 0.5 or greater, you can round up one one beat.

0:48:30 if it's less than 0.5 you can round down, okay? So the heartbeat for this rhythm is going to be 79 beats per minute. Now, saying that, remember we call that rhythm regular. In order to use this

9 Calculating Heart Rate – Regular Rhythms (revisited)

0:48:43 – 0:49:38



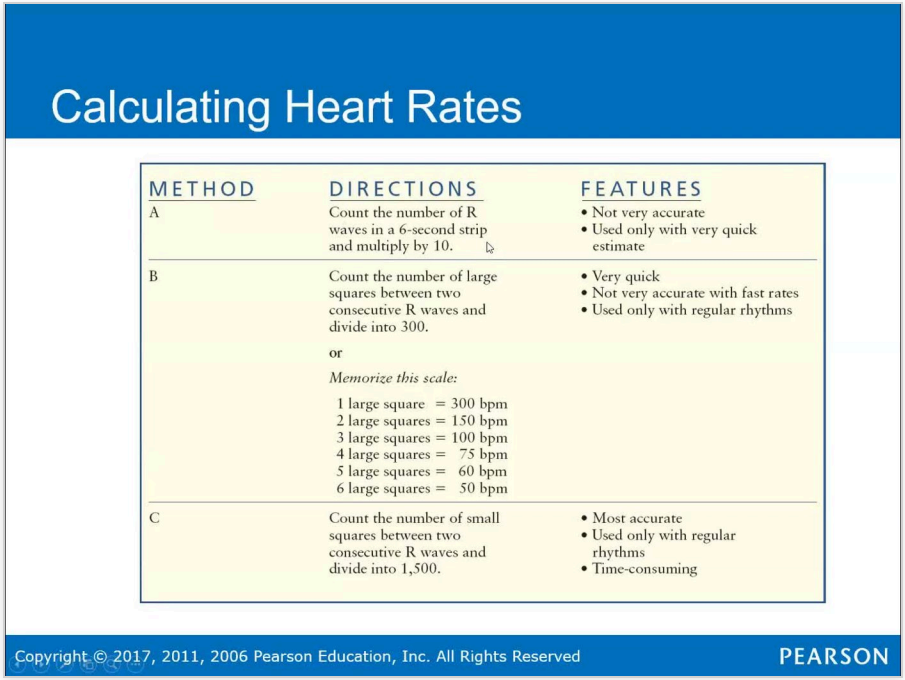
0:48:45 method here, counting the number of small squares between our waves and dividing it into 1500, we can only use this method when R to R is regular. So the only time we're going to use this rhythm is when the underlying rhythm is regular.

0:49:05 We cannot use this method when R to R is irregular because the number of small boxes between our ways varies so much, it won't give us an accurate heart rate for the strip.

0:49:27 So what do we do if the rhythm is irregular? Like this rhythm here, we determine that this rhythm is irregular. So what do we do? So if we look at the top of our slide here, when the rhythm is irregular, we're going

10 Calculating Heart Rates: Methods

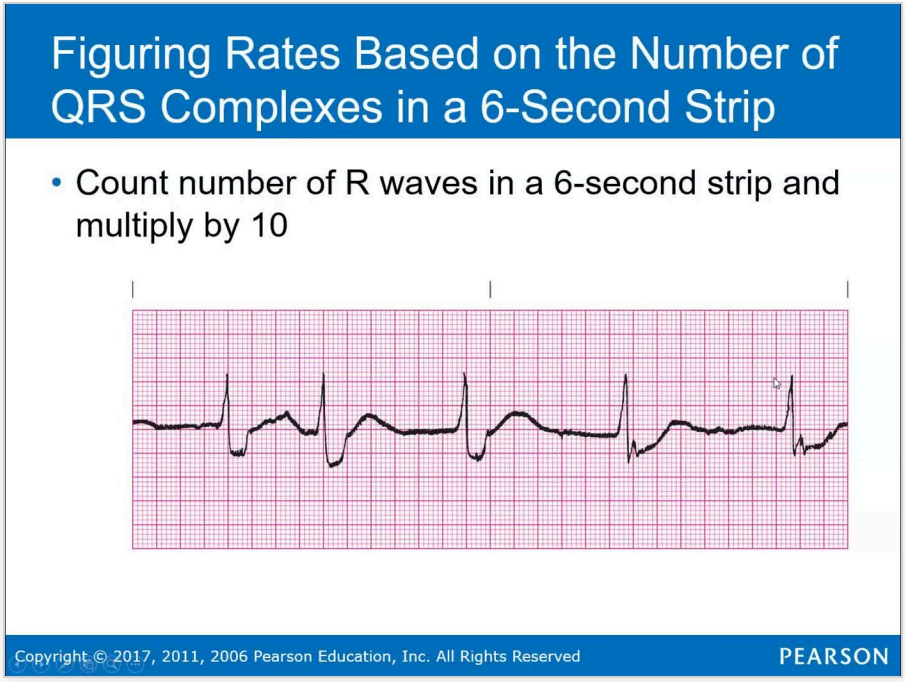
0:49:38 – 0:49:53



0:49:45 to count the number of our waves in a six second strip and multiply it by 10. So one, two, three,

11 Figuring Rates in a 6-Second Strip (revisited)

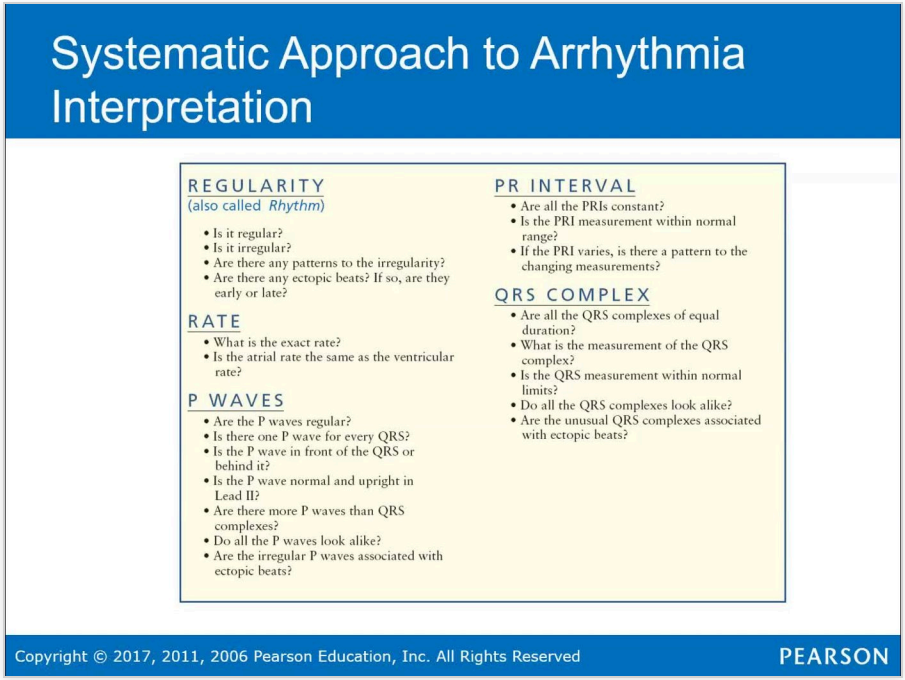
0:49:53 – 0:50:33



0:49:56 four, five, the heart rate for this strip is 50. Everybody found those are the only two methods we're going to use to count the heart rate in this class. When the rhythm is regular, when the rhythm is regular, interrupted by one or two or three air topics, we're going to use this method. When the rhythm is irregular, we're going to count the number of R waves, multiply it by 10, and that's going to be our heart rate. So the next thing we're going to evaluate is the

12 Systematic Approach to Arrhythmia Interpretation (revisited)

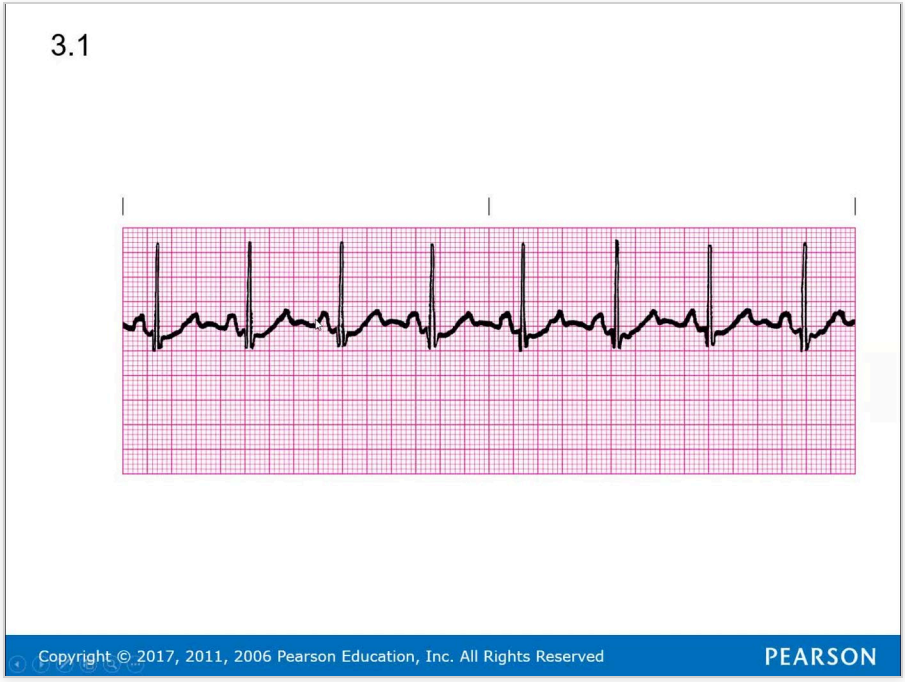
0:50:33 – 0:50:59



0:50:41 the P-waves regular. A regular P-wave is positively deflected and it looks like a dome shape.

13 Practice Strip 3.1

0:50:59 – 0:51:47

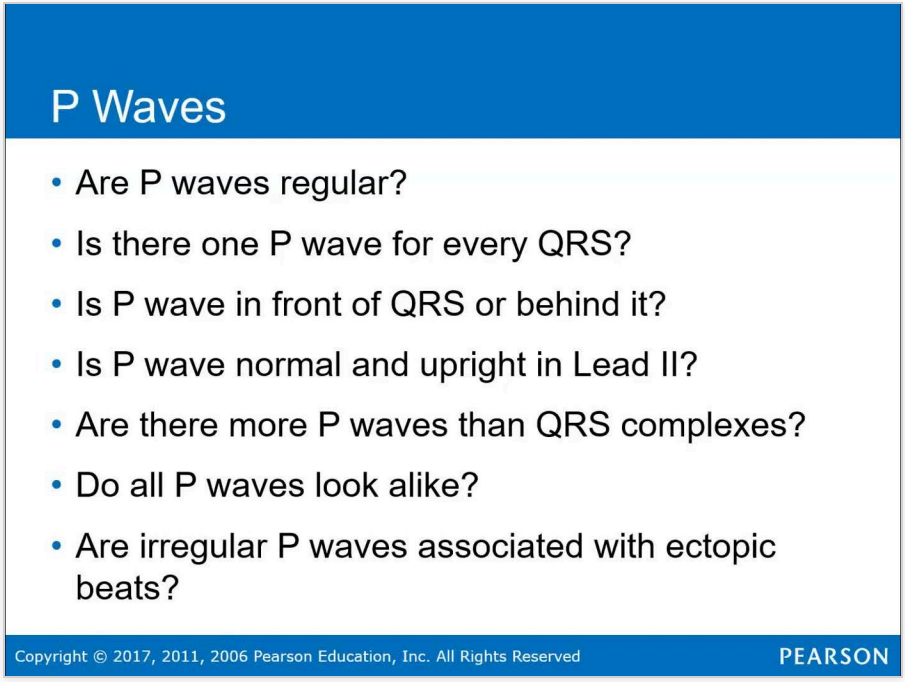


0:51:00 So this P-wave and then this P-wave and then this P-wave. So they're fairly rounded at the top. And then when I look at the shape, which the book calls morphology, morphology means shape, more importantly, the morphology, If the shape is the same, it tells us that that P-wave is coming from the same pacemaker every time.

0:51:30 So when we do get these ectopic beats that I'm referring to, the P-wave for that beat is going to look different than the P-waves from the sinus node. And that's one of the ways we're going to tell it apart, okay? So are the P-waves regular? So are they upright? Do they look the same? Typically, there's only one P wave in front of each QRS.

14 P Waves

0:51:47 – 0:53:23

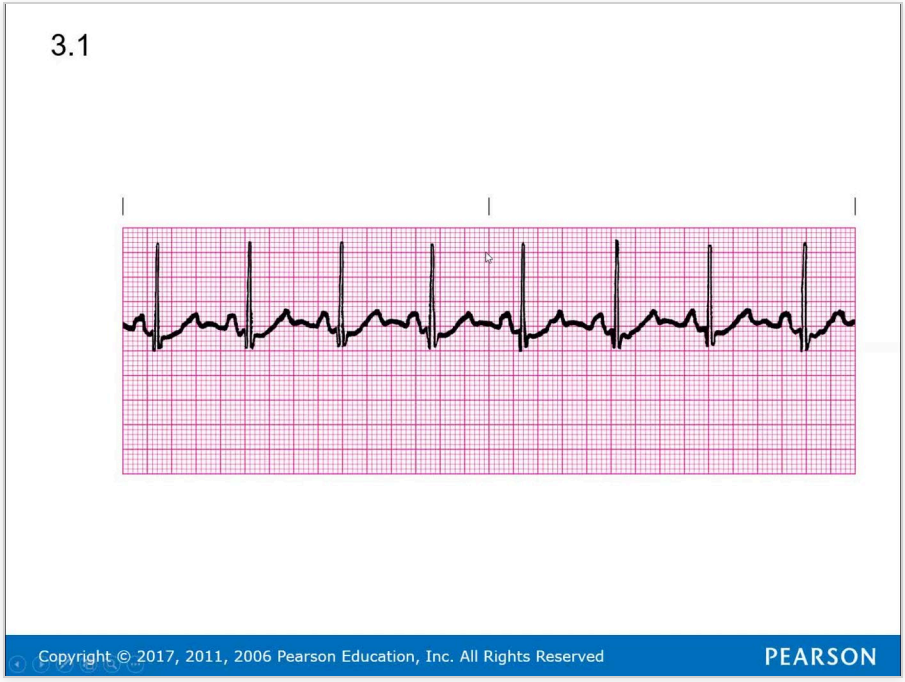


0:51:58 Are you guys comfortable when I say P wave in front of the QRS? Meaning the P wave, I see the P wave on the strip before I see the QRS. And the P wave should come before the QRS because we want the heart to beat from the top down.

0:52:20 there are there are some odd times where the P wave can actually fall behind the QRS. We'll discuss that when we get to junctional rhythms. And some other questions here, are there more P waves than QRS? Do the P waves look alike? And then again, I already said, is there P waves that look different than the rest of the P waves? That would be an ectopic beat. You guys comfortable with the word ectopic. I talked about this two weeks ago. An ectopic beat usually refers to an irregular beat. Ectopic beats usually come from irritability. Irritability usually produces premature beats. What do I mean by a premature beat is that it becomes irritable and it fires early. So when I go, if I go back to this rhythm that I've been referring to, the

15 Practice Strip 3.1: P Waves

0:53:23 – 0:55:00

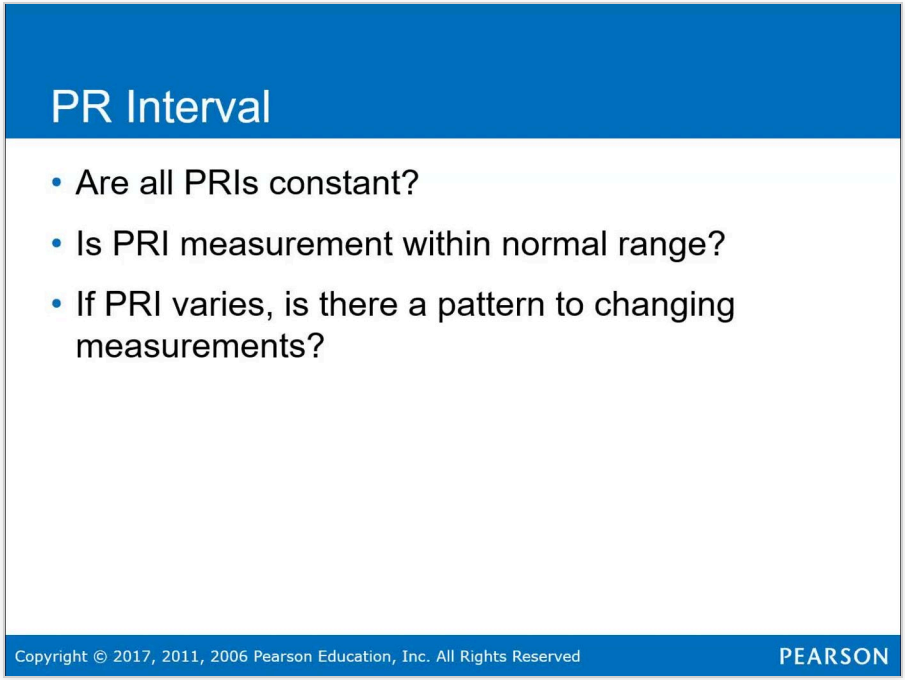


0:53:25 normal for the SA node is about 19 boxes and it fires every time. Well, if we have a spot in the AV node that got irritable and it fired before the 19th box, that beat might occur right here, right? So what we're gonna see is the spacing is going to be early. That's why we call it early. The normal interval is 19 boxes away. But if I get a beat that falls 10 boxes away, that beat 10 boxes is earlier than 19 boxes. So early beats, premature beats usually result of irritability. We can have irritability from atrial cells, so we can have premature atrial complexes. We can have irritable cells in the junction. We can have premature junctional contractions, and we can have irritable spots in the ventricle, which we call premature ventricular contractions. Okay, we'll get to those when we cover each one of those chapters. It's not something you have to worry about now, I'm just trying to familiarize yourselves with the terminology. So part of this is going to be learning the content. And then other

0:54:34 part of this is going to be understanding the terminology. Okay, so we talked about regularity, we talked about rate, we talked about P waves, we got to talk about PR interval.

16 PR Interval

0:55:00 – 0:55:20

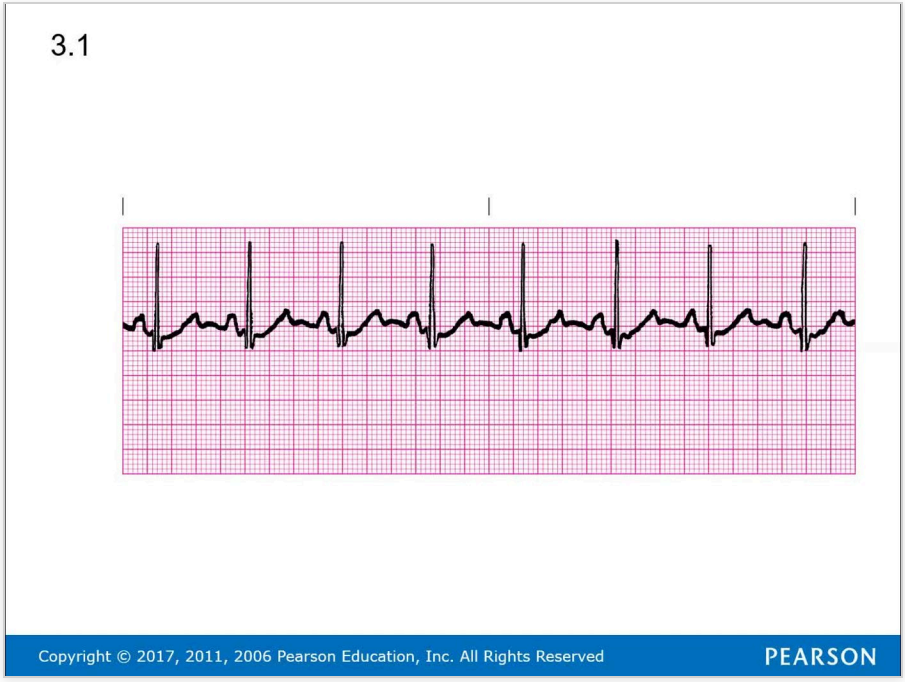


0:55:01 So what we're going to do is we're going to measure the PR interval across the strip. And we want to make sure that the PR interval remains the same across the strip.

0:55:16 So if I go back to that rhythm that I've been referring to

17 Practice Strip 3.1: PR Interval

0:55:20 – 0:56:30

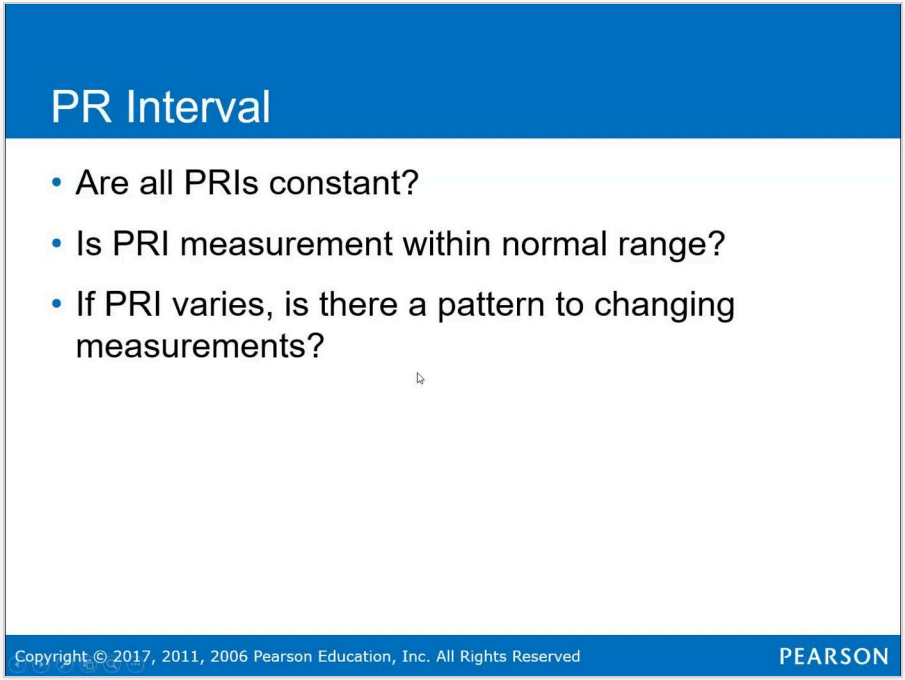


0:55:20 and we measure this PR interval, as I can do this on the screen. So if I look at this first one, the P wave starts right here on this second small line and the Q wave starts right there. So I'm gonna count from here to here. So that's one small box, two small boxes, three small boxes, four small boxes.

0:55:50 one, two, three, four. One, two, three, four. So I'm pretty convinced that this PR interval is four small boxes, and four small boxes equals 0.16 seconds. Okay? And again, normal PR interval, 0.12 to 0.2, so this fits into the normal PR interval. If the PRI varies, there may be something changing

18 PR Interval (revisited)

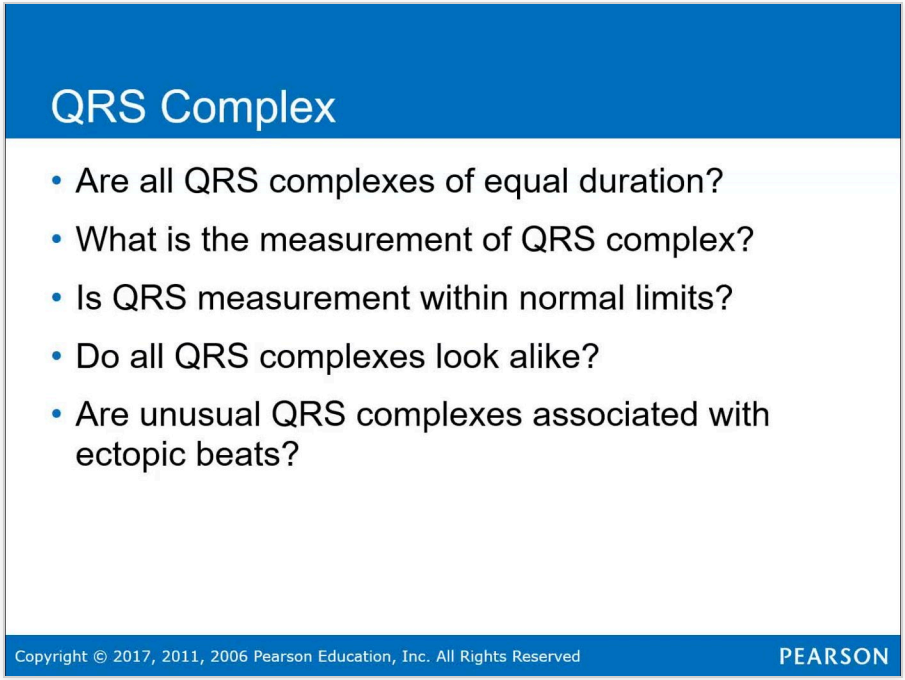
0:56:30 – 0:56:58



0:56:38 in the rhythm. It might be a different beat, or it might be a heart block. The PR interval shouldn't change that much across the rhythm strip for the majority of rhythms that we're to be discussing. And then the last thing that we need to measure is the width of the QRS complex.

19 QRS Complex

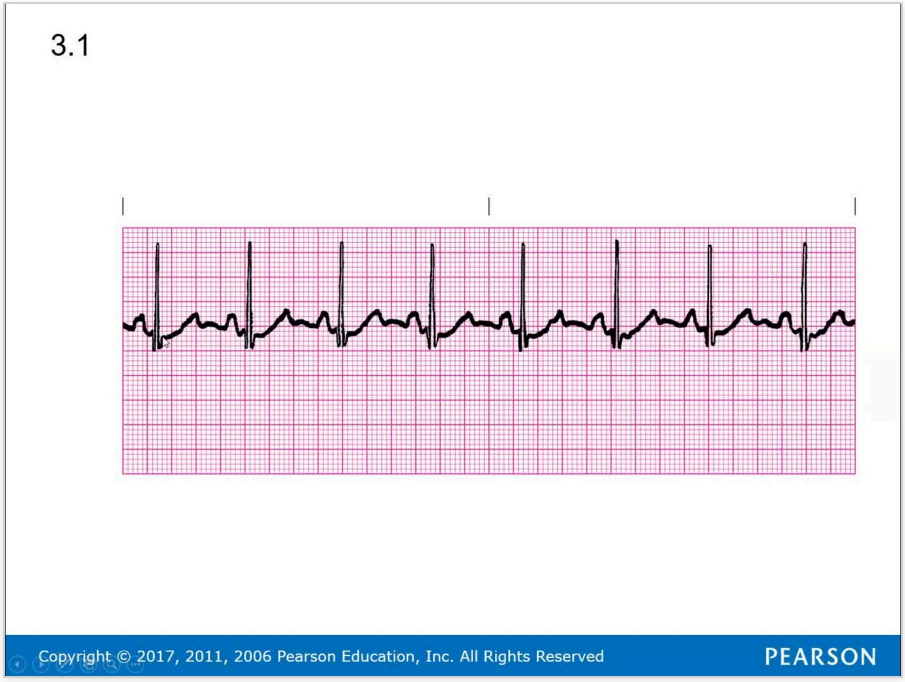
0:56:58 – 0:57:25



0:57:03 So are all of the QRS complexes of equal duration? So we're going to measure it. What is the measurement? Is that measurement within normal limits? And do the QRS's look alike? And if we have ectopic beats, the QRS might slightly vary. So let me go to that rhythm strip again and measure

20 Practice Strip 3.1: QRS Complex

0:57:25 – 0:58:56

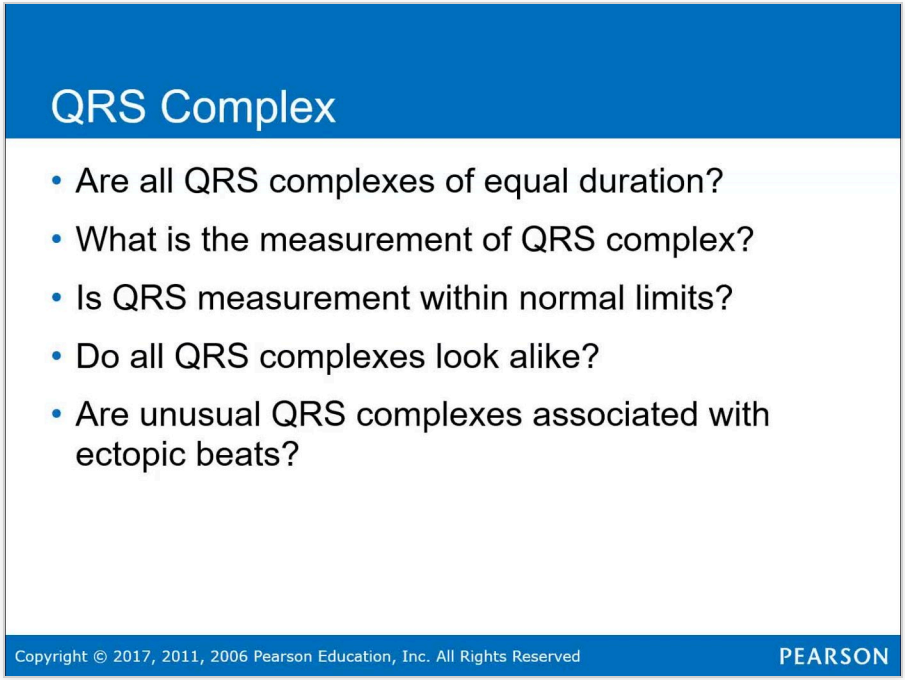


0:57:28 my QRS. So it starts on that line and it ends on that line. One, two small boxes. One, two small boxes. You see that the EKG sometimes can actually cloud your vision. So that's why I say sometimes if you take a blank piece of paper and you mark where it starts and you mark where it ends and then you bring it down to the of small boxes. And then the other thing that you can do with that that I didn't show you before. So if I want to do my PR interval, I do this and I do this, I get my four small boxes, then what I can do is I can take that piece of paper and march it across and make sure that that piece, it fits the same characteristics all the way across. And then what I can do is measure my QRS, I can put a line there for that. And I can put a line where it stops. I know that that's two small boxes. And

0:58:37 then what I can do is mark my lines all the way across to make sure that my QRS, the width of my QRS stays consistent. Just trying to guys show you guys some clues to make things a little bit easier. Okay. So those are the five pieces of information that we are going to gather, regularity, heart rate, P waves, PRI, and QRS.

21 QRS Complex (revisited)

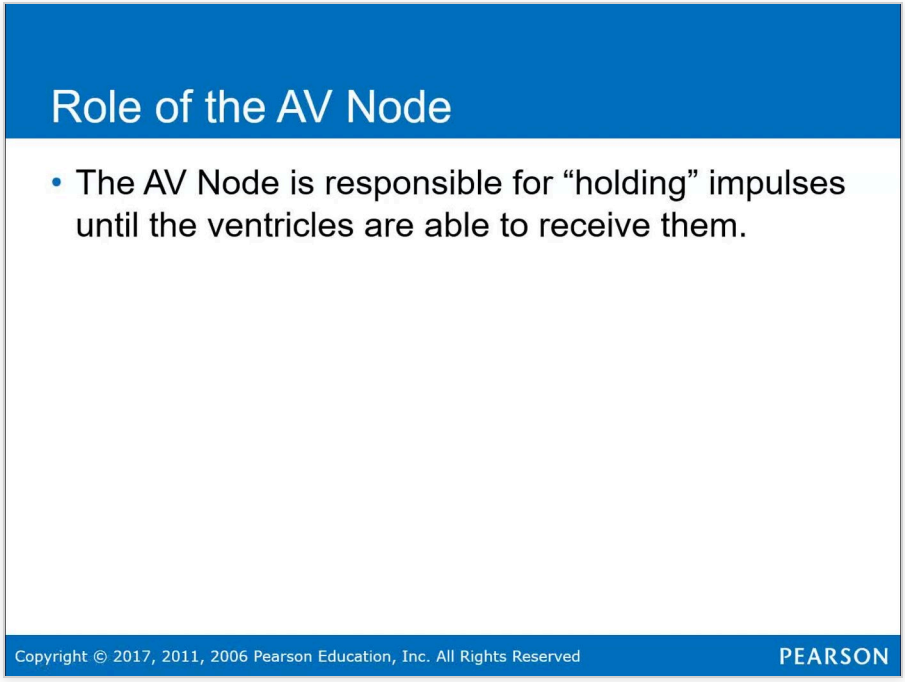
0:58:56 – 0:59:12



0:59:08 Okay, talked about the role of the AV node is responsible for holding that impulse.

22 Role of the AV Node

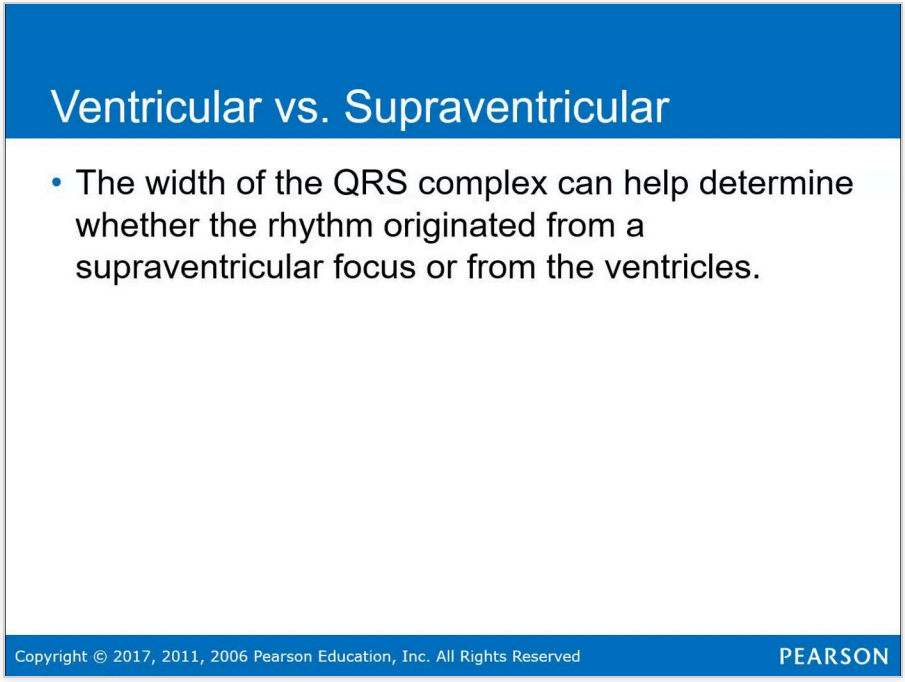
0:59:12 – 0:59:32



0:59:20 Allow the atria to contract before it sends it down to the ventricles and maximizes stroke volume. talked about that. This is an important concept, so the width of the QRS complex

23 Ventricular vs. Supraventricular

0:59:32 – 1:00:35

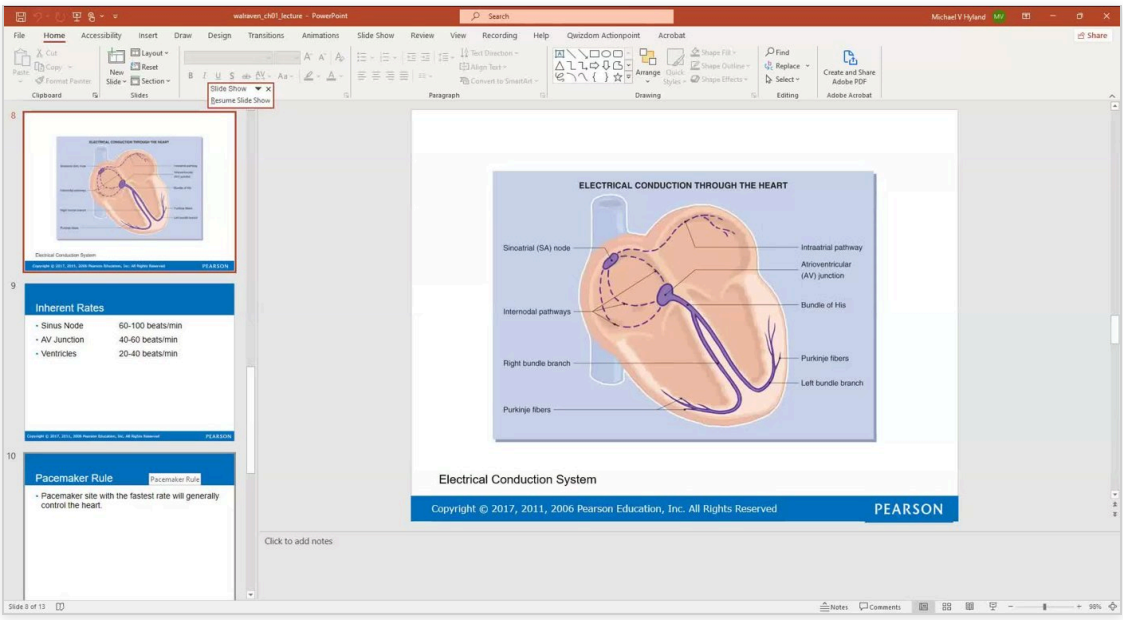


0:59:42 can help us determine where the rhythm comes from. Not always, but it's a pretty good indicator of where the rhythm comes from. I'm gonna stop share for a second

1:00:09 here. And I'm going to go back to chapter one for a second. And I'm going to call up this slide.

24 Electrical Conduction System (PowerPoint editor view)

1:00:35 – 1:05:40



1:00:38 So in that slide that I was just on, in chapter three, it had the word supraventricular. The prefix supra means what? Coming from above. So when I say supraventricular, it means that that beat, the origin of that beat came from above the ventricles. So hear me out for a second. When we look at the heart's conduction system, anything that originates from the SA node or an atrial muscle cell or the AB node itself, in order for that impulse

1:01:43 to make the ventricles contract, that impulse has to go through the bundle branches to the Purkinje fibers. When an impulse travels on a nerve, does it travel fast or slow?

1:02:02 Remember that nerve impulses, nerves are usually insulated with myelin sheath and they have nodes of reinere where it actually can jump and skip along the nerve. So when a impulse travels on a nerve, it travels very fast.

1:02:23 So any time we get an impulse that comes from above the ventricle, it has to be delivered to the ventricular muscle cells through the bundle branches and Purkinje fibers, which means the impulse comes down and is delivered quickly and contraction occurs quickly. In fact, contraction occurs quick enough that when we measure it, it's 0.12 seconds or less on the paper.

1:02:53 So when the QRS is 0.12 seconds or less, we call that a narrow or normal QRS. This is a super important concept, normal. So super ventricular. So remember that EKG, what is an EKG? it's electrical recording over time. So we're recording this electrical activity over time from left to right. If conduction or contraction occurs quickly, the waveform will be more narrow. If it takes longer for the narrative impulse to spread the electrical activity, it's gonna record that

waveform on paper wider because it travels slower. that should make sense, right? So if we have a beat that comes from above the ventricles, the only way that that impulse gets to the ventricles for ventricular contraction is if it goes through the bundle branches to the Purkinje fibers.

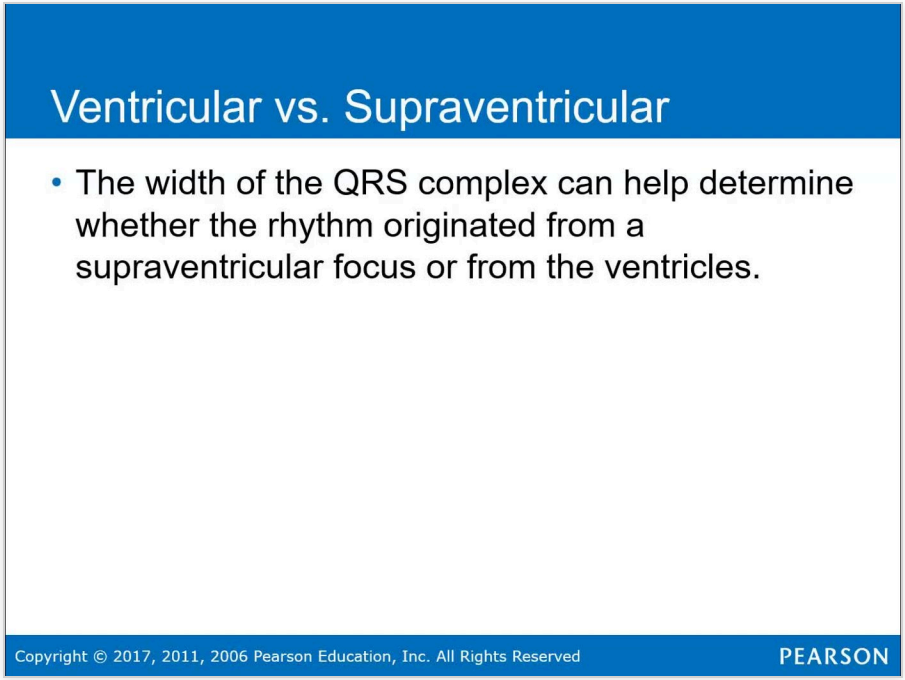
1:04:07 It travels quick, conduction occurs quickly or contraction occurs quickly. And so we're gonna get a narrow QRS, 0.12 seconds or less. and I called that a narrow QRS or a normal QRS.

1:04:28 I will typically use the word narrow. If we have an impulse that starts over here, we get a spot in the left ventricle that becomes irritable and it fires and it causes the ventricles to contract. There's no nerves for it to travel across. that impulse has to spread cell to cell all the way across. So the impulse spreads slowly. So when we record that QRS on paper, it's gonna be a wide QRS or greater than 0.12 seconds. So it's important that you understand the concept between a narrow QRS and a wide QRS.

1:05:39 So back to this supraventricular, so supraventricular beats can come from the SA node, an atrial cell, anywhere in the atria, or the AV node itself.

25 Ventricular vs. Supraventricular (revisited)

1:05:40 – 1:06:57

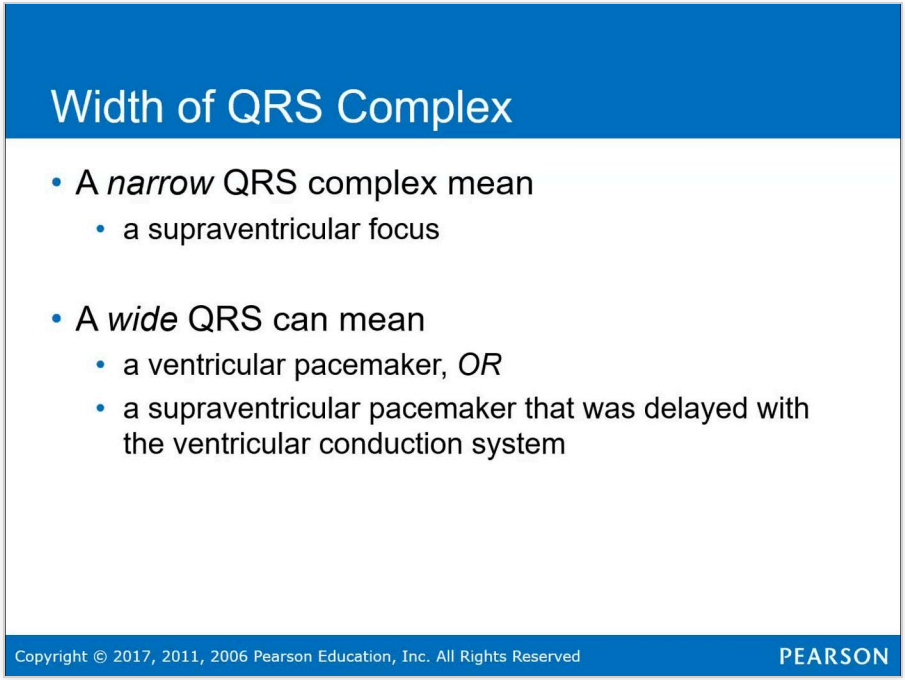


1:05:55 And I might remind you that the bundle of his is actually part of the AV node. So superventricular means coming from above. So that beat, when it has a narrow QRS, that is not a ventricular beat in origin. It is a superventricular beat in origin,

1:06:19 meaning that it originated from above the ventricles and contracted the ventricles quickly without a narrow or a normal QRS, okay? So that is gonna be one of the clues that we're going to use to determine, right, if we have a wide QRS most of the time it's not coming from all of those pacemakers that I just mentioned, it's probably coming from somewhere in the ventricles. It's gathering a clue to identify where it's coming from and it can be a single beat or it can be an entire rhythm. So narrow QRS and you might even hear me say narrow complex tachycardia.

26 Width of QRS Complex

1:06:57 – 1:10:33



1:07:07 Another way of saying a narrow complex tachycardia would be a superventricular tachycardia because narrow complexes come from above the ventricles. Above the ventricles is superventricular. Same thing. A wide QRS a lot of times is going to be a ventricular pacemaker.

1:07:26 Now this is confusing, very confusing, but we can have a pacemaker, a superventricular pacemaker, that fires a beat and ultimately produces a wide QRS. Where this is confusing is, when we get a superventricular pacemaker that produces a wide QRS, it means there's a slowdown in one of the bundle branches. So that impulse slows down and it takes longer to deliver it to the ventricular muscle cells. So even if it's a super ventricular pacemaker, it has a wide QRS. I'm not going to give you any of those at any point in time throughout the semester. Too confusing to address in this period of time. That's the ones that look like tombstones, right?

1:08:24 No. The ones that look like tombstones are the ST elevations of somebody having a heart attack. that has nothing to do with the QRS itself. Those are ST elevations. So the S wave comes down, here's the isoelectric line, the S wave comes down, but then it connects right to the T wave and it looks like a tombstone that's right next to the QRS. Those are generally not a good sign, and that's why we call them tombstones. Do you see the ventricular pain taker delay like that in like patients that have issues with, you know, their violent chief, like MS patients or? Typically not from that type of thing.

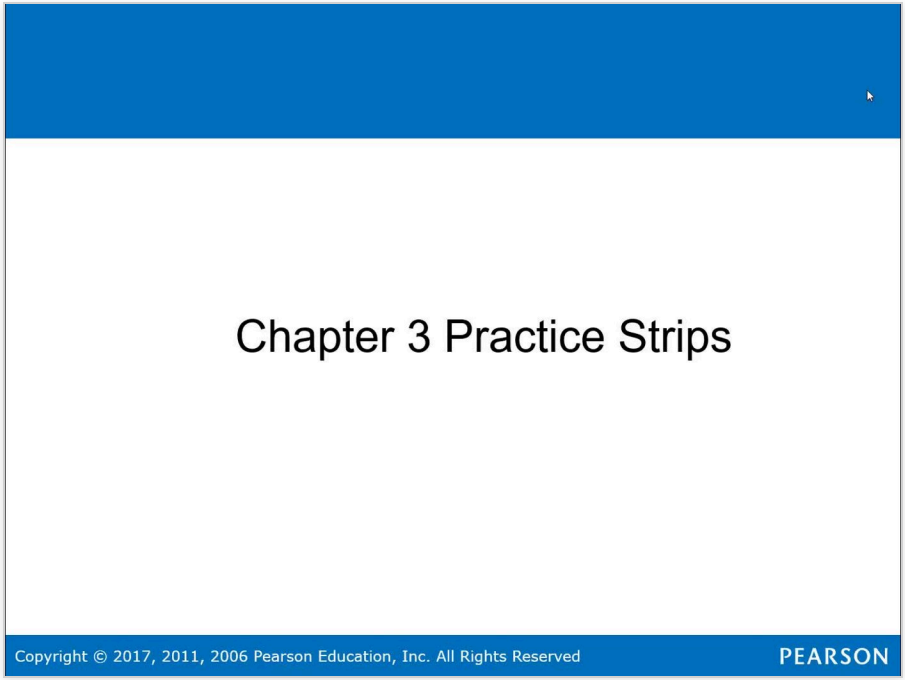
1:09:09 Typically these are patients that have cardiac disease. Patients that have coronary artery disease, coronary artery blockage, patients that sat, that's had an MI in the past.

1:09:22 Viral infections can affect nerve impulse transmissions. There's a lot of different things that can cause it. You could have an electrolyte imbalance, right? Because remember, in order for a cell to depolarize, right, the cell has to be, you have to meet the threshold of the cell with a stimulus. Sodium channels open, sodium rushes in, and then the potassium channels open and the potassium leaks out. Well, what if you got a high level of potassium in your blood? Remember these electrolytes flow from a high concentration to a low concentration. It's not a lot of salt inside the cell. Most of the salt sits outside the cell.

1:10:03 So when the sodium channels open, they rush in. The majority of the potassium sits inside the cell, not outside the cell. So when the potassium channels open, it has to go from a high concentration inside the cell to a low concentration outside the cell. But if you have a lot of potassium outside the cell, the potassium doesn't leak out and you don't depolarize. So a lot of different things that can cause that, okay?

27 Chapter 3 Practice Strips

1:10:33 – 1:12:28

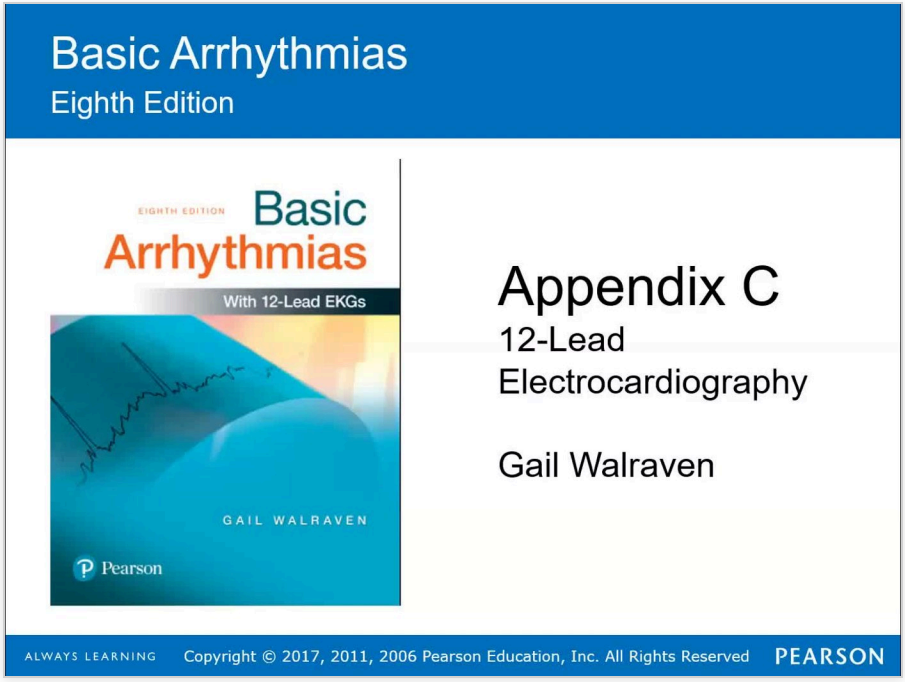


1:10:35 Okay, why don't we take a couple minute break and then I'm gonna go over a short PowerPoint for 12 lead EKGs and I'm gonna let you go. And then the rest 101 people are gonna stay. Did everybody put their homework in the pile up here? please and thank you. Let's take a five minute break. I didn't like the other one. For you guys, I'm recording these lectures. And if you didn't, I don't know if you noticed, but like for you, I recorded week one. And then what I do is I put the link to the recording back in week one. So if you want to go back and watch it, I did, I have not edited it yet. And it didn't record as I was going through the course outline. So there is like 15 minutes of no voice, no recording at all.

1:11:54 And just the course outline that's on the front. So you just got to fast forward that until we get to the lecture piece. But I'm also, once this downloads, I'll put this recording in week two as well. So you can go back and listen. Okay. So that's why I'm recording. All right. So

28 Appendix C: 12-Lead Electrocardiography (title slide)

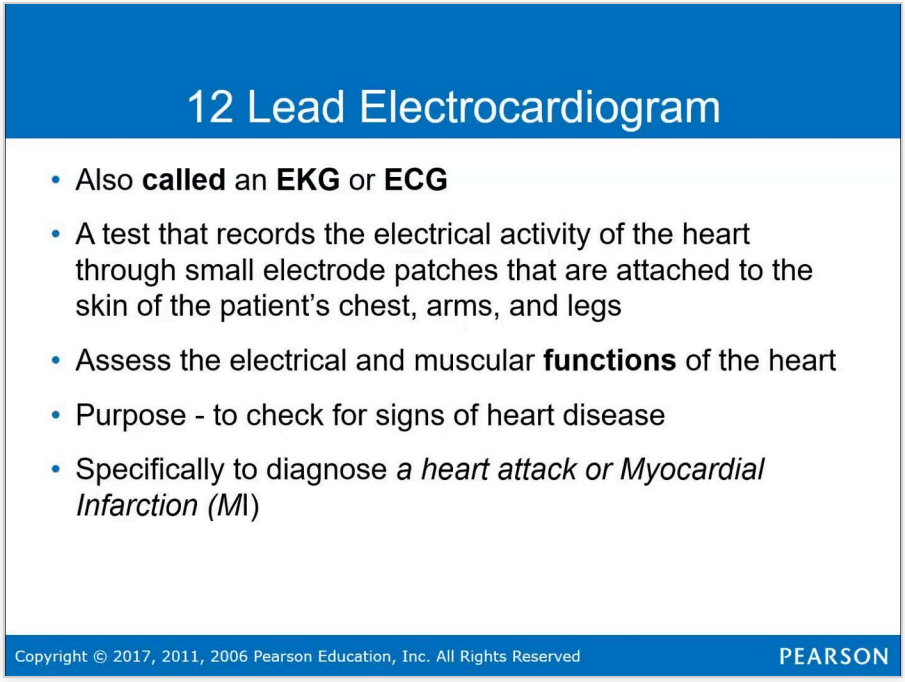
1:12:28 – 1:12:52



1:12:30 12 lead EKGs. So I wanted to introduce a little bit about 12 lead EKGs. And I'm going to go quickly, because what I want to do is call up a couple of strips at the end of that previous PowerPoint and go through some information gathering, just so you get a little bit more comfortable with that. Okay, so I'm going to go through this fairly quickly. All right, so 12 leads. So 12 lead

29 12 Lead Electrocardiogram

1:12:52 – 1:13:32

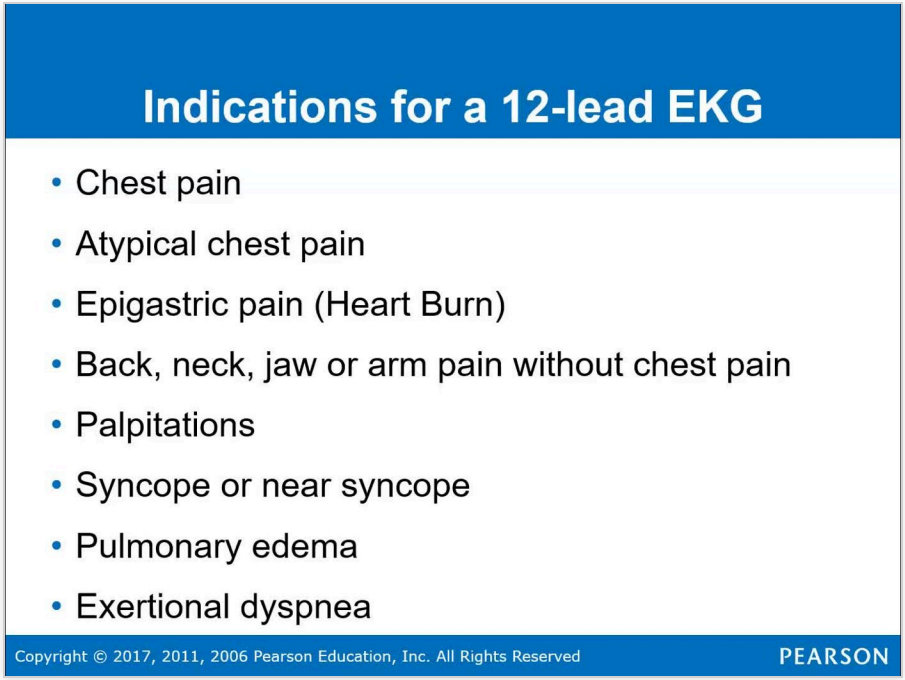


1:12:56 yet electrocardiogram, also called EKG, ECG sometimes, a test that records the electrical activity of the heart through small electrode patches that are attached to the skin, the patient's chest, arms and legs, assesses the electrical and muscular function of the heart. So there is a little bit of muscular function that it can assess.

1:13:19 We wanna check for signs of heart disease, specifically the biggest indication is to diagnose somebody having a heart attack or something called the myocardial infarction, MI. What are

30 Indications for a 12-lead EKG

1:13:32 – 1:19:19



1:13:34 the indications to do with 12 leads? So somebody's complaining of chest pain. So what happens? You get blockage in one of your coronary arteries and that blockage is usually some kind of fatty deposit that gets laid down on the inner lumen or the inside of one of the coronary blood vessels. remember I said that the coronary blood vessels are the blood vessels that come off of the root of the aorta and they're on the outside of the heart muscle and they go down and feed all of the cardiac muscle with oxygen-enriched blood. If you lay down a fat deposit inside one of those coronary arteries, it reduces the blood flow to the section of the heart that that artery feeds with oxygen rich blood and sometimes that blockage can get bigger and bigger or sometimes when you lay down a fatty deposit inside the blood vessel it can harden and then once it gets hard it can crack and some of it actually can break off and then float downstream inside the coronary

1:14:49 artery and then it gets wedged inside the coronary artery and now it starts to block blood flow. Your body then thinks that's a disruption in the in the blood vessel wall and so what your body does is you start to form a blood clot on that piece of plaque that's wedged in the coronary artery so your body starts to clump platelets on there right and as you clump more platelets on there, it continues to block the blood flow more and more. The more platelets that you clump or aggregate on that, the more blood flow that gets blocked. So the distal part of the heart muscle that that vessel is no longer feeding as much or no blood flow at all now is not getting oxygen. And so cells that don't because cellular metabolism becomes altered when you don't have oxygen.

1:15:55 So you go from aerobic cellular metabolism, which is cellular metabolism with oxygen, to anaerobic cellular metabolism, which is cellular metabolism without oxygen. And so that portion of the heart muscle will actually cause this patient some pain. and we call that pain chest pain, or the more medical term is something called angina, A-N-G-I-N-A. Some people will say angina, and some people will say angina. Potato, potato, we'll stop it there. Okay, so somebody comes in complaining of chest pain, and sometimes, sometimes it's not just chest pain, sometimes it doesn't feel like chest pain to that person, it might feel like a crushing pain, or it might feel like indigestion, like, you know, like you ate a lot of spicy foods, and now all of a sudden you got this heartburn.

1:17:08 So sometimes it just feels like heartburn. Sometimes they can have jaw pain. Sometimes they can have back pain, back pain that radiates to the leg or to an arm. So a lot of different presentations. Sometimes the people don't have pain at all, which we call a silent MRI, okay?

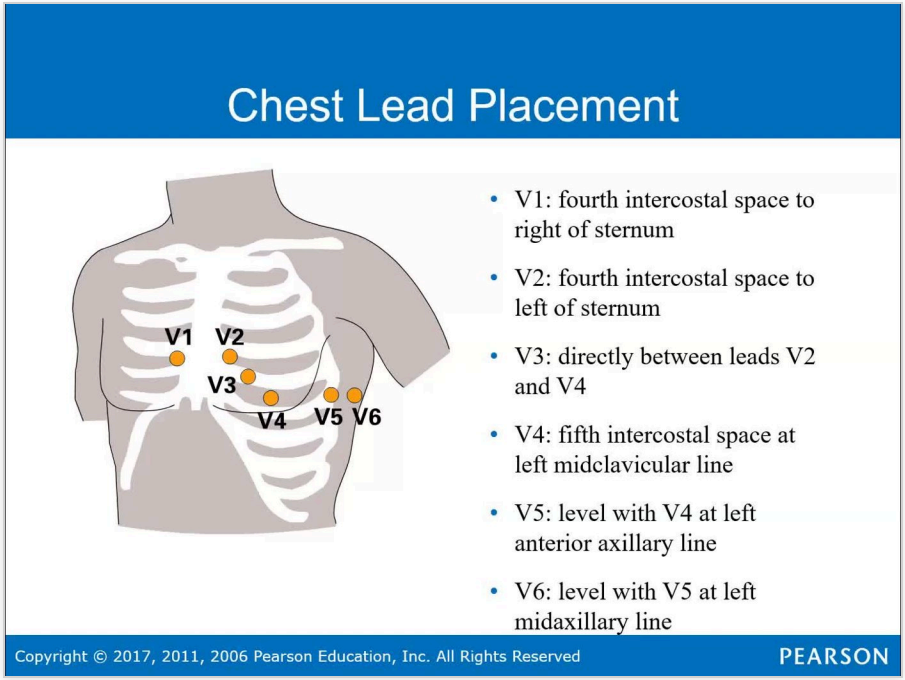
1:17:29 So chest pain, which may be not a normal chest pain that they've experienced in the past. That's what I mean by atypical. Or maybe they confuse that with heartburn, back, neck, back, neck, jaw, arm pain. And you don't necessarily have to have chest pain when you have back pain or arm pain or jaw pain. It might feel like their heart's fluttering a little bit, something that we call palpitations.

1:18:01 They might have an episode of what we call syncope or near syncope. Syncope is when you pass out. If this heart attack is bad enough and it affects the left ventricle and the left ventricle is the chamber that is mainly responsible for pumping blood out to the body, if it affects the left ventricle's ability to pump blood and move it forward, the blood doesn't go forward, it goes backwards. It backs up into the lung and you get lot of fluid in your lung and we call that pulmonary edema. It's like when your sump pump overflows at home because your pump's not working well, the water stays in the bin and it doesn't get pumped out. And then the other thing that can happen with this is, you know, when they sit and they don't move around, it calms down. They get up and start walking around and they start exerting themselves, now all of these symptoms come back again.

1:19:05 Okay, so lead placement, we talked about lead placement. We talked about how we want to get the limb leads on the limbs, not on the trunk. But if we have to, we can go on the trunk. We

31 Chest Lead Placement

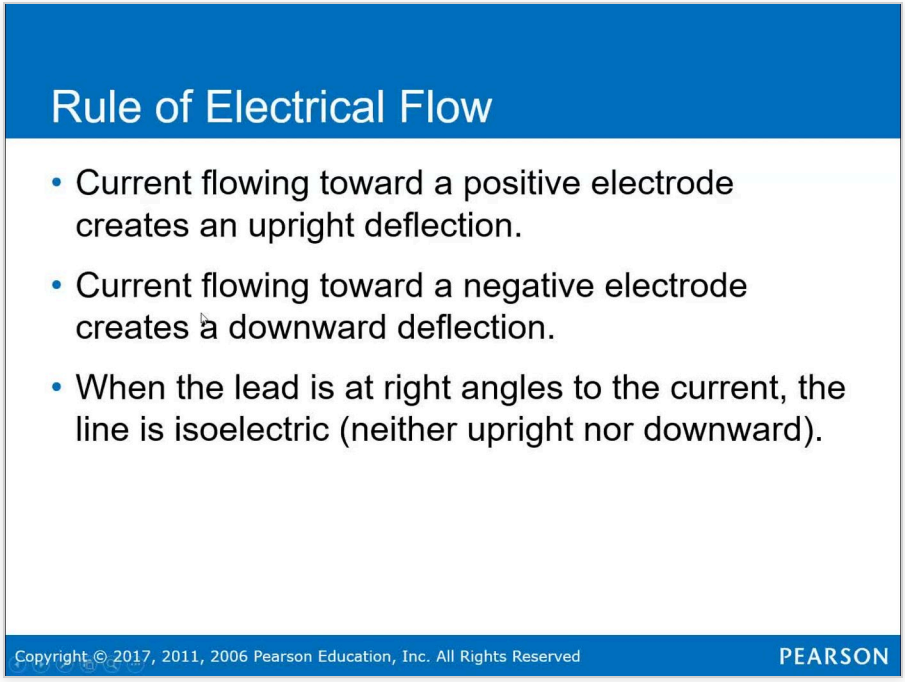
1:19:19 – 1:19:33



1:19:20 talked about V1, 2, 3, 4, 5, 6. So for those of you that are going to clinical might want to have an idea of where these leads get placed before you go to clinical. We talked about rule

32 Rule of Electrical Flow

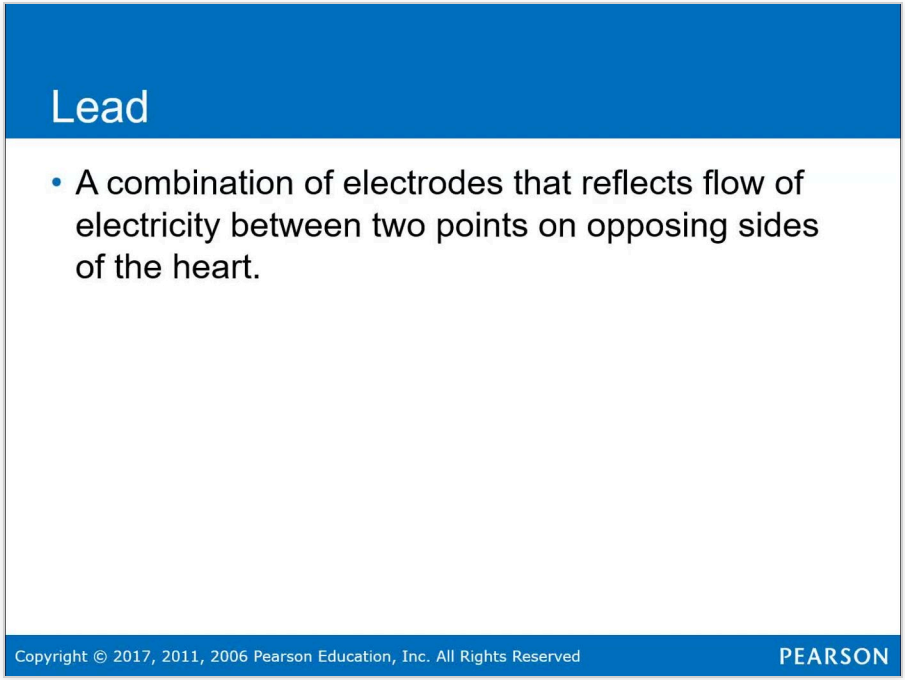
1:19:33 – 1:19:47



1:19:36 electrical flow, so I'm not going to go through this again. That's positive or negative. We talked about what a lead is. It's a combination of electrodes on opposite side of the heart

33 Lead

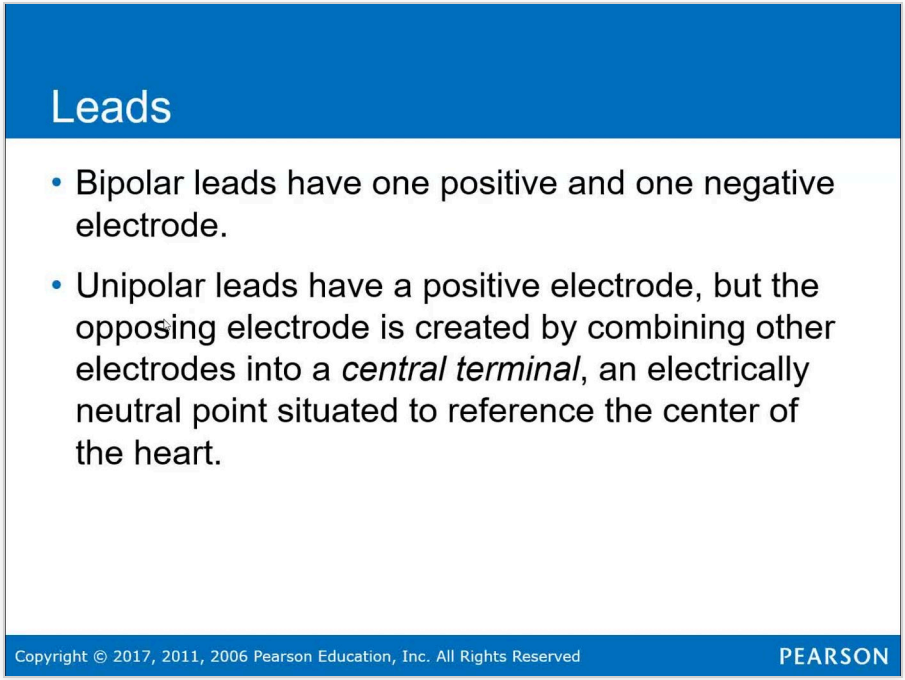
1:19:47 – 1:20:04



1:19:53 that reflects the flow of electricity between those two points. And we talked about how we can see 12 different leads. Ten electrodes on the patient, but we get

34 Leads: Bipolar and Unipolar

1:20:04 – 1:20:35

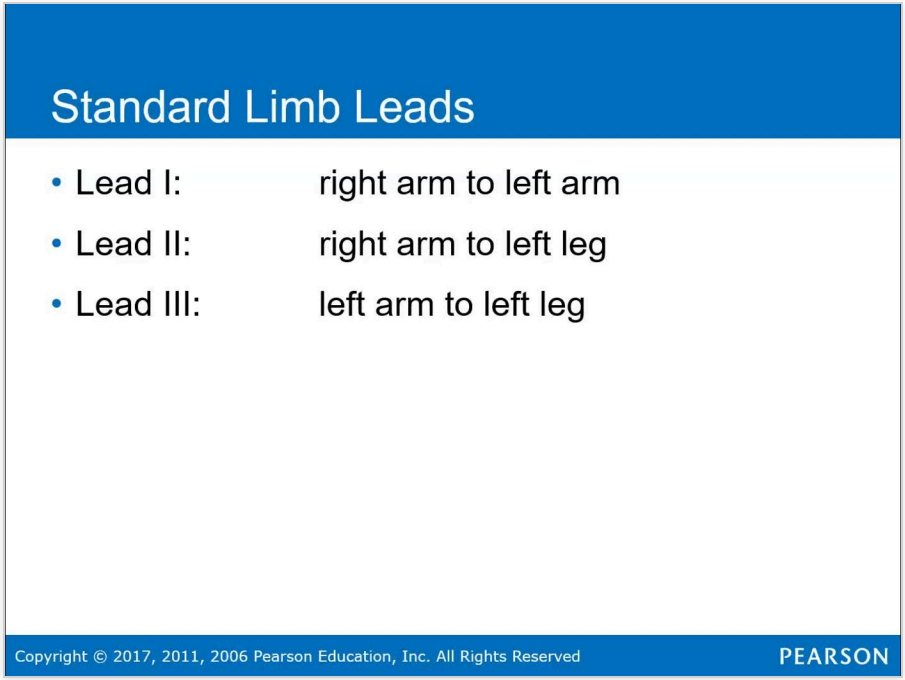


1:20:05 to see 12 different leads. Bipolar leads, obviously the opposite ends. One is negative, one is positive. Unipolar usually just has a positive electrode. Not something super important. Those are the augmented leads.

1:20:29 Talked about the horizontal and the frontal plane, which leads cover that surface of the heart.

35 Standard Limb Leads

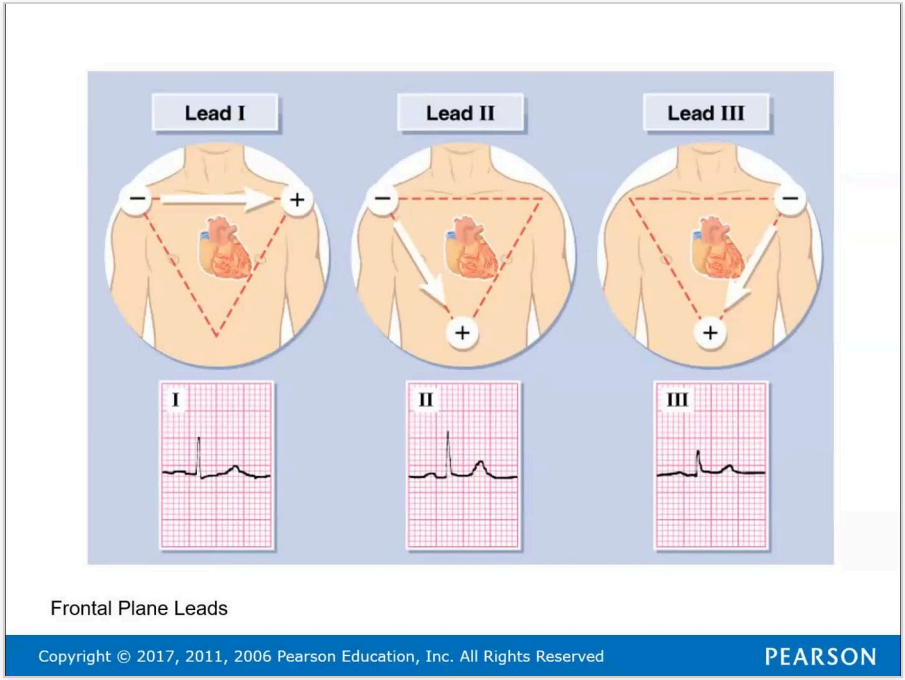
1:20:35 – 1:21:18



1:20:37 Talked about how lead one is right arm to left arm, lead two is right arm to left leg and lead three is left arm to left leg. We talked about how the common most common monitoring lead is lead two. We want to monitor going straight across the top of the heart. We talked about this lead one, two, three. I'm told it's a triangle. So okay,

36 Augmented Leads

1:21:18 – 1:27:09



1:21:25 One of the reasons why we do a 12 lead EKG is because we want to get, we want, we want to have a lead, a print out of each lead that covers every surface of the heart.

1:21:44 Okay, so I talked about some of the surfaces of the heart last two weeks ago, sorry two weeks ago, right. I talked about how the heart sits in the chest, little bit more to the left side, it's anterior. I talked about the bottom of the heart being the inferior wall. This side of the heart is lateral wall. This side of the heart is anterior wall. And then this side of the heart is the posterior wall. And then we also talked about the septal wall, right? And so we know that these coronary arteries come off the aorta, and each of those coronary arteries are responsible for providing blood flow to all of those different surfaces of the heart.

1:22:29 So we want all of these 12 different electrical views because if somebody has coronary artery disease, it can potentially alter the flow of electricity across that surface of the heart. So what we're looking for is do we have ST changes or changes in the Q wave or ST depression, right? Where are we seeing these electrical abnormalities? In what lead are we seeing it? We know what lead that we know that that lead covers this surface of the heart. And then we know that this coronary artery provides blood flow to that surface. So when we get a good idea of where we see these disturbances in what lead, we know what surface of the heart is, we know what coronary arteries get fed there.

1:23:24 Okay, typically what's going to happen is somebody comes in the ED complaining of chest pain or angina, and they're short of breath, we're going to put them on oxygen, they may give them some kind of blood thinner, like aspirin or something like that, they may give them nitroglycerin, which dilates the coronary arteries.

1:23:43 and maybe a little something like morphine to relieve the pain and morphine also causes a little basal dilation, which actually reduces the preload and decreases the work on the heart because if the heart is still working hard, remember, you know, when you're working out your muscles require more oxygen. So what we want to do is make the heart relax so it doesn't have to be so hard, right? because the harder it beats, the more oxygen it needs. The more oxygen it needs, now we have a big gap between what's being supplied and what it demands, and that's when we have a heart attack.

1:24:22 That should make sense, okay? All right, so lead one, again, is right arm, right shoulder to left shoulder, or right arm to left arm, and that is going to cover the superior surface. Lead two is going to cover the anterior surface going from right arm to left leg and you see these little snippets. We're going to get these little snippets so we're going to get this is what lead one looks like this is what lead two looks like and then left arm left leg lead three right so for example, lead two is going to go right across the anterior portion of the heart. Lead three is going to cover the lateral wall. So we go here to here, this is all going to cover the lateral wall. So if we have ST changes, in lead three, we know that that's the lateral surface of the heart, right? And we know that somewhere on the left coronary artery, usually the the left anterior descending or the circumflex

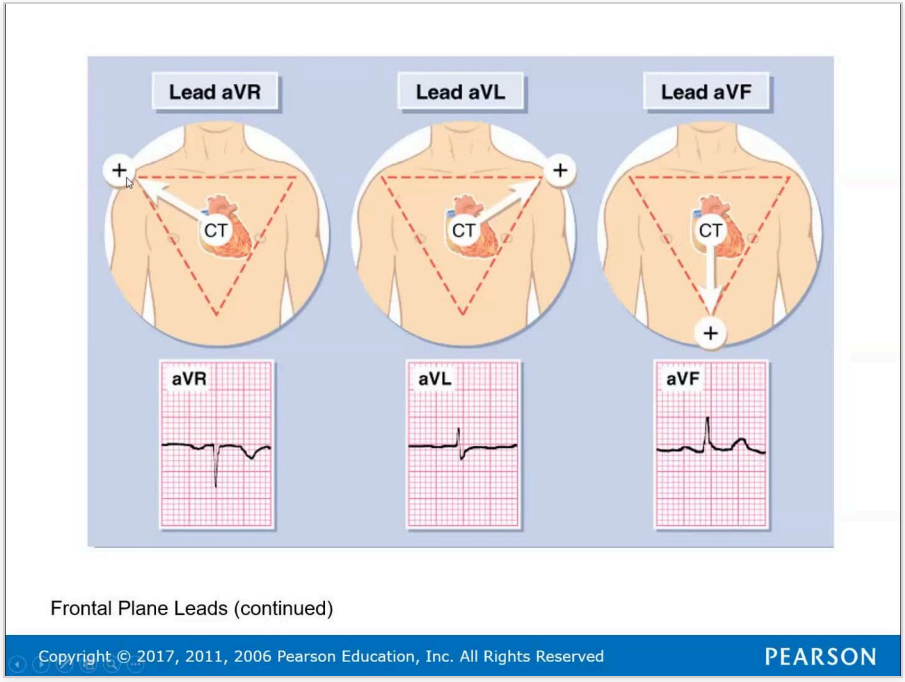
1:25:34 is gonna feed that wall, that surface of the heart, okay? So somebody comes in complaining of chest pain, we put them on oxygen, give them some morphine, maybe give them some aspirin or another type of blood thinner, right? So that they're, why do we need the blood thinner? Cause I was talking about how the platelets clump, right? Or right, the aggregator clump. So we wanna give them a medication so that the plate would stop clumping onto that blockage or that clock and further getting bigger and blocking more oxygen, right? To that cardiac muscle cell, right? And then if the 12 lead shows something significant, right?

1:26:14 Then we got to send them up to the calf lab. They got to go for a cardiac calf. And then in the calf lab, they're going to squirt dye into the coronary arteries. They're going to track this dye and see where the blockages are. And if the blockages are soft or hard, if the blockages are soft, they can stent the artery, put a stent in it right up in the cath lab. If the blockages are hard and calcified, you can't put a stent in that, they're gonna have to go for open heart surgery and have a coronary artery bypass grafting, all right? Not that you guys need to know all of that, but that's kind of the logical progression of how things would work, okay?

1:26:56 All right, so AVR, so we're gonna get these small snippets with AVR, AVL, AVF, central terminal to the right, central terminal to the left, and central terminal to the left leg. So again, we're going to get these snippets and look at AVR,

37 Frontal Plane Leads (continued)

1:27:09 – 1:27:58

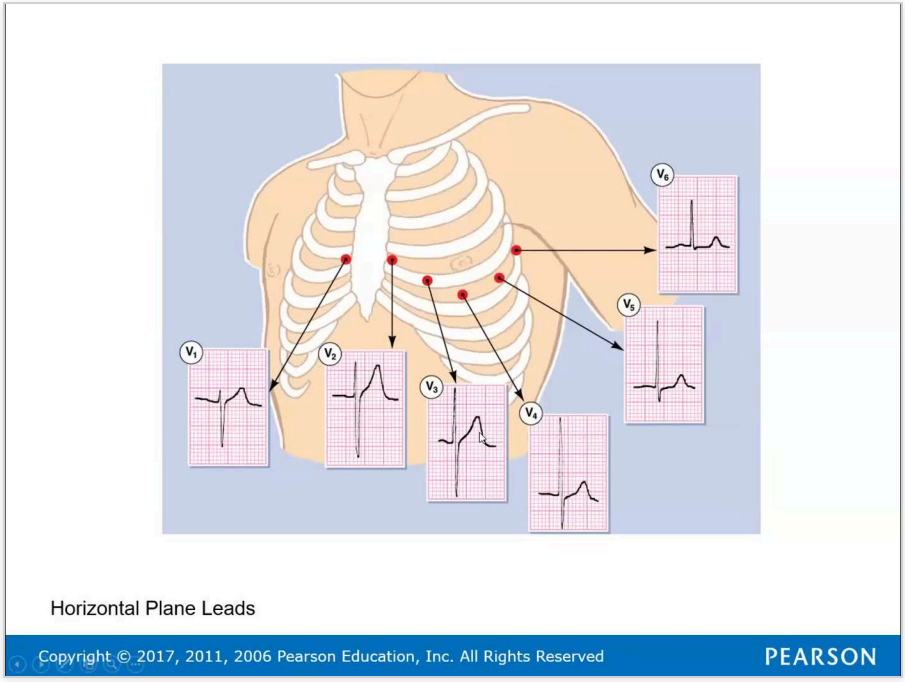


1:27:15 central terminal to the right arm. The right arm is a negative electrode. So look at all of our waveforms. They're upside down. They're inverted. That's normal in AVR. Okay, we're going to look

1:27:29 at AVL and this is going to be a unipolar lead so this is 90 degrees to that so half of it's going to be positive half of it's going to be negative again not that you need to know that right and then AVL left leg is a positive electrode everything is positive so what we're looking for in these little snippets is do we have changes the end of the S wave through the T wave Okay? And then the pre-corial lead.

38 Horizontal Plane Leads

1:27:58 – 1:29:27

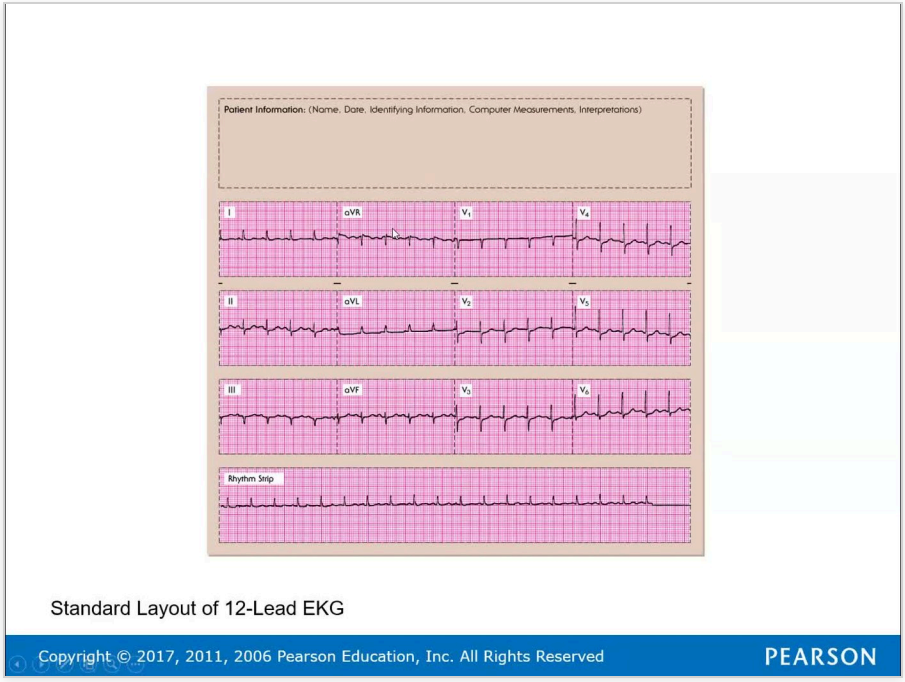


1:28:00 So we're going to get a little snippet of V1, V2, V3, V4, V5, and V6. Okay? And if you look at this, right, so specifically like V3, V4, right? So remember, the heart sits in the middle of the chest right here. And I said that this is the anterior wall. the anterior wall, most of that interior wall is the left ventricle. So if you see those ST changes in V3 and V4, right? That means that the anterior wall of the heart has been affected. That's the left ventricle, right? Those are the bad MI. So somebody comes in with ST elevations and tombstones in V3 and V4, that's an anterior wall MI.

1:28:50 Those are the ones where, right? and that's the left coronary artery or the left anterior descending coronary artery. Usually that refers to major blockages in those areas. That person's got to get to the cath lab right away. And if you go to a hospital that they don't have that cath lab, then the alternative is they have to be given a drug that breaks up the clot, something called TPA, tissue plasminogen activator, which breaks up blood clots, okay? Again, you don't have to know that.

39 Standard Layout of 12-Lead EKG

1:29:27 – 1:33:00



1:29:27 And then this is what a 12 lead looks like. So we get this small snippet of lead one, two, three, AVR, AVL, AVF, and then V one, two, three, V four, five, and six. And then we'll get a six second strip on the bottom, which is usually lead two, okay? So that's what a 12 lead looks like. All of the patient information will get typed up in here. So when you guys go to clinical and do a 12 lead EKG, they actually have a scanner now, right on the 12 lead. So you scan the patient's bracelet and it'll populate all the patient's information. And they'll show you how to do that when you go to clinic.

1:30:03 Okay. All right. Questions on that? We do have a test. So when we come back and have the test in two weeks, there will be 40 multiple choice questions on the test. And then there will be five EK, six second EKG strips that will be on the test. And I'm not going to expect you to interpret the rhythm at this point. What I am going to expect when I give you those five rhythm strips is that you gather the five pieces of information you tell me and so what what's going to happen is is it's it there's going to be an EKG strip and below that I'm going to have typed out regularity and there'll be a space for you to fill in is this rhythm regular or irregular you're going to count the number of spaces between our ways and to tell me whether this rhythm is regular If the spaces are the same, you're going to write regular for regularity, right?

1:31:23 If the spaces are not the same, you're going to write irregular. Then underneath regularity will be the heart rate. So I'm going to expect you to be able to count the heart rate. And if the rhythm is regular, you're going to divide the number between small boxes into 1500. If the

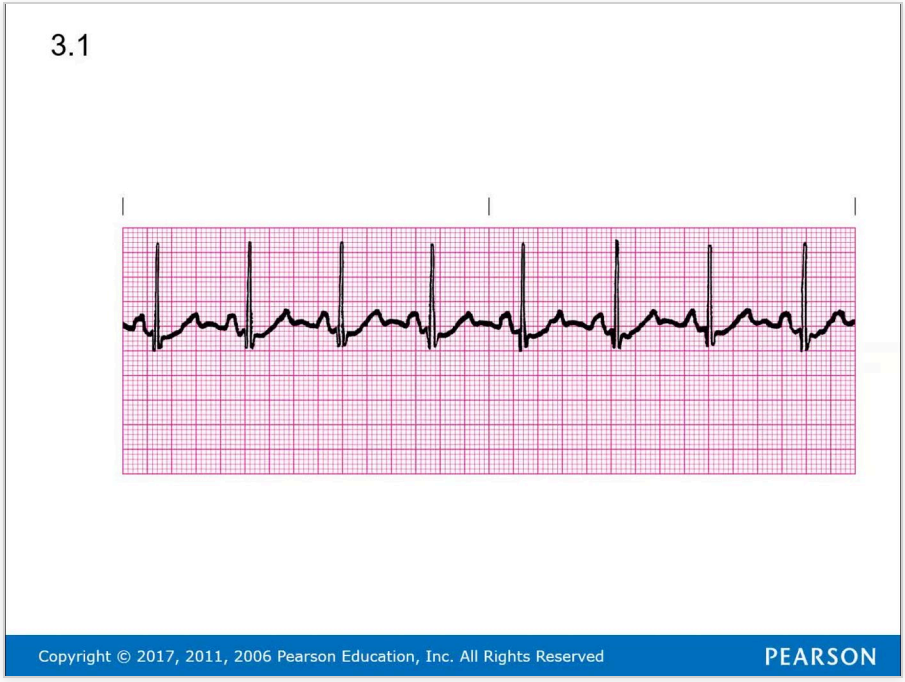
rhythm is not regular, you're going to count the number of R waves and multiply by 10.

1:31:45 For the P-waves, you're just going to tell me you're going to describe the appearance, right? And that's one of the things I want to do is go through one or two strips to tell you what I expect you to write, okay? So then you're going to have to do P-waves, you're going to have to measure the PRI and measure the width of the QRS, and each one of those will be worth one point. So each of those strips will be worth five points, regularity, rate, P waves, PRI, QRS, and each of the multiple choice questions. So it'll be 40 of those. And so I would focus on all of the stuff that we covered in week one and today.

1:32:24 I'm probably only gonna have one or two questions about 12 lead EKGs on the tests. So I would not focus on a lot on 12 leads, okay? All right, let me get back to chapter three.

40 Practice Strip 3.1

1:33:00 – 1:43:43



1:33:00 Okay, so again, we're gonna have five pieces of information. So I'm gonna have an EKG strip here, just dab on. Underneath, it's gonna say regularity. And it's actually, if you look at the homework,

1:33:42 you look at today's homework, you're gonna see, actually, I'm not gonna do this, you're gonna see what I'm expecting. but you look at today's homework.

1:33:53 As a matter of fact, how about I stop share here and we'll go through the first trip on the homework. I'll give you a bonus.

1:34:06 Let me just call up homework number two. Okay, I'm not gonna review the questions. You guys are gonna do the fill in the blanks, but what I am gonna do is start with number 11.

1:35:22 Okay, so let's count the number of small squares We'll do the first one out loud. You guys can count with me. So 5, 10, 15, 16, 17, 18, and I think there's two halves on each end. Let me actually make this bigger.

1:36:00 Where is my zoom on here? View. I know where it is. Down here. here. There we go. Okay, there we go. 5, 10, 15, 16, 17. There's a half here. 18. And there's a half. So I'm going to call this 19. 5, 10, 15, 16, 17, 18. And I'm going to call that a half and that a half. So that's 19, 5, 10, 15, 16, 17, 18, 19. I guess I technically have to call that a half. So I'm going to call that 19 and a half.

1:36:49 5, 10, 15, 16, 17, 18, 19. Now we'll call it 20, 5, 10, 15, and 4 is 19 and a half is 19 and a half. 5, 10, 4 is 14, 4 more is 18, and 2 halves make 19, 5, 10, 15, 18, and a half. So you can see we got a little bit of a spread here. We go from 18 and a half up to 20. That's one and a half boxes. Not enough to call this irregular. So this rhythm is regular.

1:37:39 Did everybody follow me on that? So under regularity, you're writing regular. We're going to do our heart rate now. What was the most? What did we have? We had, what, 19? So I'm saying, what were we? We had 19, 19.5, 20.

1:38:06 So what's the majority? Is it 20 or 19.5? 19.5. So 1,500 divided by 19.5. All right, now I'll do 1,500 divided by 19. Yeah, 79. Okay. So our heart rate is 79. P-waves. So again, I can't see the P-wave on this first QRS, but I can see it here. P-wave QRST, P-wave QRST. So if I look at my P-waves, there's the first one, the second one, the third one, the fourth one, the fifth one, sixth one, seventh one. So they are all upright, positively deflected. So I'm going to put a U for upright. So you can just put U for the first part.

1:39:09 So I'm looking for you to tell me whether they are positively deflected, which would be represented as upright. And then do they all look the same?

1:39:20 So do all of these P waves look the same? So is the shape exactly the same? Yeah, don't split hairs on me. They all look the same. So I would call them uniform and appearance.

1:39:35 So what you're going to do is put U slash U upright and uniform, or you can put U and U upright and uniform. All right, now we've got to measure the PRI.

1:40:09 So again, we're going to measure from the beginning of the P wave right here to the beginning of the QRS, which starts halfway in between. So I'm going to count one, two, three, four and a half small boxes. Again, I said something about halfsies and wholesies. So a half of a small box would be how much time? 0.02, right? So if I have four small boxes, that's 0.16 plus another half a box. That's going to be 0.18. So this PRI is four and a half small boxes for 0.18 seconds.

1:41:00 Everybody follow that. All right, now we're gonna measure the QRS. And before I measured the QRS, you know, I already looked at this, so we should be measuring those PRIs across the strip, not just the first one. And if I took a piece of paper and measured and knocked out my PRI, you can see literally that's four and a half small boxes of them down.

1:41:33 Right? And so if I do that, then I do that, that's four and a half, that's four and a half, that's four and a half, that's pretty consistent across the strip.

1:41:47 all right now we're going to measure the QRS all right so we're going to measure from here right here to right here here to right there how many small boxes i actually think it's one and a half now here's the thing is i'm going to give you one small box grace period. So if I think this

is one and a half, my answer would be .06. If you think it's two small boxes and you put .08, that will be within one small box of what I had, I won't mark that wrong. So I'm going to give you one small box grace period on the PRI and the QRS. All right, So you should be able to take it from here, okay?

1:43:07 All right, so I am available via email. So as you're going through this homework, if you have questions, send me an email. Remember, I'm typically in clinical with the respiratory care seniors. Otherwise, I will see you guys in two weeks. We will have a, you're gonna get one hour to take the test and then we will have a lecture after the test.

1:43:32 Okay, for those of you that are in rest 101, if you just will stay seated for a second.